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KEESLER AIR FORCE BASE (AETC)

KEESLER AIR FORCE BASE
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Weather

WEATHER SUPPORT

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This instruction implements Air Force Policy Directive (DAFPD) 15-1, *Air Force Weather Operations*, Air Force Manual (AFMAN) 15-111, *Surface Weather Observations,* and DAFMAN 15-129, *Air and Space Weather Operations*. It establishes responsibilities and weather support procedures, and provides general information for weather services, including weather forecasts, observations, advisories, watches, warnings, and reciprocal support. This instruction applies to assigned, attached and/or associate units supported by the 81st Training Wing (81 TRW) at Keesler AFB, MS and units assigned or attached, including Air Force Reserve, except where noted otherwise. This instruction does not apply to ANG. Ensure all records generated as a result of processes prescribed in this publication adhere to AFI 33-322, *Records Management and Information Governance Program*, and are disposed in accordance with the Air Force Records Disposition Schedule, which is located in the Air Force Records Information Management System. Refer recommended changes and questions about this publication to the Office of Primary Responsibility (OPR) using the DAF Form 847, *Recommendation for Change of Publication;* route DAF Form 847s from the field through the appropriate functional chain of command. **Attachment 1** is a glossary of references and supporting information. Compliance with the attachments in this publication is mandatory. The use of the name or mark of any specific manufacturer, commercial product, commodity, or service in this publication does not imply endorsement by the Air Force.

SUMMARY OF CHANGES

This publication has been rewritten and needs to be completely reviewed. Changes to this instruction were made to better align Keesler guidance with Air Force guidance and requirements,

most notably pertaining to the recently updated DAFMAN 15-129, *Air and Space Weather Operations*.

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Chapter 1

GENERAL INFORMATION

1.1. General. A team comprised of 6 contractors forms the Keesler AFB Operational Support Weather Support Office (BOS/WSO). In conjunction with the 26th Operational Weather Squadron located at Barksdale AFB, Louisiana, they provide weather support to the 81 TRW and associate units assigned or attached to Keesler AFB, Mississippi. Basic concepts and procedures are outlined in Air Force and Air Education and Training Command (AETC) directives, instructions and manuals. This document establishes requirements and procedures for areas of weather support that must be coordinated at the local level to meet mission needs. It consolidates weather support requirements and procedures for garrison operations but does not cover procedures for emergency war operations or other special operations/procedures that are covered in applicable base plans and instructions.

1.2. Concept of Operations.

1.2.1. According to DAFMAN 15-129, *Air and Space Weather Operations,* Air Force Weather (AFW) developed an integrated weather support structure in order to ensure weather operations were conducted at the strategic, operational, and tactical levels, and provide more meaningful environmental information, products, and data to operational customers. According to the CSAF-approved Strategic Plan for AFW Reengineering (8 Aug 97), the 26 OWS will provide resource protection and terminal aerodrome forecasts, (when BOS/WSO is not available) 26 OWS will also provide regional, and operational-level weather products and information, meteorological watch (METWATCH), and Pilot-to-Metro Service (PMSV) (currently only available through phone patch), and limited flight weather briefing support for Air Force and Army locations within designated geographic regions.

1.2.2. According to the Strategic Plan, the 26 OWS at Barksdale AFB (BAFB), Louisiana will provide regional and operational-level weather products and information to Air Force and Army units in the Southern region of the Continental United States and for Air Force and Army forces operating in the Southern area of responsibility (AOR).

1.2.3. Based on guidance from the Strategic Plan, weather flights at each Air Force and Army location will be reengineered to focus on providing tactical-level weather products and information needed for mission execution and provide weather products and information to the commander of deployed forces during wartime, contingency, and exercise operations.

1.3. Responsibilities. General responsibilities of the 26 OWS and BOS/WSO are outlined in DAFMAN 15-129, Specific responsibilities of the 26 OWS and the Keesler's BOS/WSO are defined in the datasheet between BOS/WSO and 26 OWS and in this instruction. The 26 OWS provides timely, accurate and relevant weather products as follows: weather warnings, watches, and advisories when BOS/WSO is not available. Terminal Aerodrome Forecasts (TAFs) are issued by BOS/WSO. The 26 OWS will also provide regional and operational-level weather products; tactical-level weather products; limited PMSV support during non-duty hours; and flight weather briefings to transient aircrews passing through Keesler AFB when BOS/WSO is not available. BOS/WSO will issue all advisories, watches, and warnings (during duty hours) and create tailored

weather products focused on the local mission. The 26 OWS will provide back up support (TAFs, watches/warnings/advisories, etc.) whenever the BOS/WSO is unable to complete their duties for any reason. BOS/WSO. Additionally, BOS/WSO will support the base in thoroughly educating agencies on the purpose, applicability, and operating procedures of weather products in reference to Emergency Management as outlined in DAFI 10-2501, Emergency Management Program.

1.4. Duty Priorities.

1.4.1. BOS/WSO Duty Priorities. BOS/WSO duty priorities are listed in **Table 1.1**. Forecasters will use good judgment in complying with these duty priorities, especially where there is imminent danger to life and/or property.

Table 1.1. BOS/WSO Duty Priority Listing.

1. Wartime defense of the duty site/location
2. Perform Emergency War Order Tasks (e.g., Deploy Personnel)
3. Execute Evacuation / Continuity of Operations Plan
4. Issue/Disseminate Imminent Hazardous Weather Warnings
5. Respond to Aircraft/Ground Emergencies
6. Issue/Disseminate Imminent Weather Advisories
7. Respond to PMSV calls supplementation/backup
8. Disseminate weather observation
9. Disseminate Urgent PIREPs (UUAs)
10. Disseminate Terminal Aerodrome Forecasts
11. Provide Flight Weather Briefings
12. Collaborate WPs with Supported Units
13. METWATCH/Amend Weather Products
14. Respond to Support Assistance Request (SAR) or Request for Information
15. Provide Staff Briefings / Non-Standard WPs
16. Accomplish Weather Functional Training
17. Accomplish Administrative Tasks

1.5. Operational Hours. 26 OWS Weather Forecasters are on duty 24 hours a day, seven (7) days a week. BOS/WSO is available 24 hours a day Monday through Friday. Weather Forecasters will also be available from 1000L to 1700L on Saturday and 0700L to 2300L on Sunday during the 403 WG Unit Training Assembly (UTA) weekend. 403 WG will coordinate with 81 OSF leadership and BOS/WSO LF for the times they need support. There is no BOS/WSO forecaster scheduled on federal holidays or approved airfield closures. A weather forecaster can be contacted by phone at DSN 597-3305 or COMM 228-377-3305 or by email at **81OSF.KeeslerWeather@us.af.mil** When the airfield is closed, a member of BOS/WSO is on standby to be recalled in the event the airfield is opened outside of normal operating hours or in the event of severe weather.

1.6. Assumptions, Shortfalls and Limitations.

1.6.1. 26 OWS Assumptions. Adequate resources and communications will be available to execute the Datasheet and sufficient weather intelligence will be available from various sources on which to base weather operations and production.

1.6.2. BOS/WSO Assumptions. Weather support can only be provided if the appropriate facilities, funding, communications, personnel, and indigenous support (e.g., water, power, etc.) are available.

1.6.3. BOS/WSO Shortfalls. Some services may not always be available due to BOS/WSO staffing, station evacuation, or other higher priority missions. In these cases, weather information can be obtained by contacting the 26 OWS at DSN 331-2651 for weather briefings, or DSN 331-2622 for the Keesler AFB forecaster.

1.6.4. **BOS/WSO Limitations.**

1.6.4.1. Wind Speed/Direction Observations. When the winds are from the northwest and runway sensor 22 is in use, observed wind-speed readings will be lower due to the Marina tree line, additional wind data is available on RWY sensor 04.

1.6.4.2. Lightning. Lightning may not be seen due to distance, low clouds, or poor visibility. Thunder may not be heard because of flight-line noise. The AN/FMQ-19 Automated Meteorological Observing System (AMOS) lightning detection system detects both cloud to ground lightning and “cloud flashes,” but only plots cloud to ground strikes on the range ring display.

1.6.4.3. PMSV Limitations. The PMSV frequency at Keesler is 267.4 MHz. Weather information is available during normal duty hours on frequency 267.4 MHz to provide aircrews with operational weather support. For aircraft outside the range of our PMSV system, when the station is closed, or during outages, they will have to obtain PMSV support from another agency. For phone patches to the 26 OWS, aircrews can contact DSN 331-2651 or COMM 318-529-2651. See Flight Information Handbook Section C for additional Pilot-to-Metro services.

1.6.4.4. Climatological Limitations. BOS/WSO maintains only limited historical and climatological information; therefore, requests beyond BOS/WSO records will require additional time for processing through the 14th Weather Squadron.

1.7. Alternate/Back-up Operating Location (AOL). In the event of a building evacuation due to hurricane, fire, bomb threat, etc., Airfield Services Element, Mission Weather Element and Staff Services Element will move to either the alternate operating location (Bldg. 4332, Rm W235) or to the hurricane evacuation shelter (Bldg. 4330 Wolfe Hall), as appropriate to continue operational flight weather support and BOS/WSO "eyes forward" responsibilities for the 26 OWS. The current AOL affords BOS/WSO with much the same operational capability as the primary operating location with the exception of PMSV. Upon evacuation from the primary operating location, Weather Forecasters will inform the ATC Tower and let them know we are evacuating and ask them to inform aircrews trying to contact Keesler Metro that our PMSV is out, and they will need to contact another location for assistance. Airfield Services Element and Mission Weather Elements providers will follow duty specific Standard Operating Procedures (SOPs) and BOS/WSO Evacuation Checklists (including a list of required backup equipment) and resume services at the alternate location as soon as possible. For flight safety reasons, BOS/WSO will not evacuate for exercises. See **paragraph 10.2.1** and **paragraph 10.3** for details on back-up procedures.

1.7.1. Limitations at the AOL. During supplementation/back up of the AN/FMQ-19 AMOS, observations at the AOL will be taken from the top floor of Bldg. 4332. This affords BOS/WSO an excellent view of the airfield. However, there is a restriction of the south end of the runway by trees and buildings. When BOS/WSO personnel are taking observations, they will be away from the phones for several minutes. Most BOS/WSO services/support should be provided, but will require a case-by-case assessment dependent upon communication line status, equipment status, etc. BOS/WSO will continue to keep an operational backup in the event of total communication failure. Expect some services to be somewhat degraded (weather products, pilot briefings, etc.) due to limited facilities and loss of dedicated data services, including various data (PMSV).

1.7.2. AOL contact information. Once BOS/WSO forecasters arrive at the AOL, they will contact the 26 OWS, Command Post and each of the flying squadrons and provide them with a contact number (228-377-8758) for BOS/WSO.

1.7.3. Evacuation for Hurricanes. Weather Forecasters will start 24-hour operations (if not already occurring) when the 81 TRW/CC or 81 TRW/CD orders the base into HURCON 3. When a hurricane or tropical storm is likely to affect KAFB, weather forecasters will continue working at the weather station for any storm forecast to be a category 2 hurricane or lower. For a category 4 or 5 hurricane, personnel will evacuate to D'Iberville High School when directed, and for a category 3 hurricane, the 81 TRW/CC will direct personnel to shelter in place or evacuate once all aircraft have evacuated the base and Airfield Management officially closes Keesler's airfield.

1.8. Release of Weather Information to Non-DOD Agencies and Individuals. Except as authorized by 81 TRW Public Affairs, non-DOD organizations and individuals will be referred to other government or private weather information services for weather data (e.g., local National Weather Service [NWS]). Any questions/clarifications will be coordinated through the BOS/WSO Lead Forecaster.

1.8.1. Weather requests by foreign nationals must be coordinated with AF Scientific and Technical Information officers before being fulfilled by BOS/WSO.

1.9. Post-Mission Analysis/Feedback. Per DAFMAN 15-129, units that regularly utilize weather support from BOS/WSO will provide post-mission/utilization feedback, when possible. This information will be used to ensure proper Quality Assurance (QA) and to provide a metrics database. Formal/informal feedback methods include:

1.9.1. Transient (faxed) and local weather brief feedback sheets located at the Mission Weather Element counter and flying squadrons.

1.9.2. Feedback sheets accompanying all faxed Flight Weather Briefings (FWB).

1.9.3. Email to BOS/WSO Lead Forecaster or call 228-377-3938. Feedback sheets can also be placed in boxes provided for that purpose.

1.9.4. Routing through the base mail system.

1.9.5. Face to face feedback after any briefing.

1.9.6. BOS/WSO will, in turn, utilize this data to refine their mission support role and gauge unit strengths and weaknesses. Representative sample sizes are necessary to accurately reflect the satisfaction with and accuracy of the mission weather support being provided to Keesler's flying units by BOS/WSO. Since the Supervisor of Flying (SOF)/Duty Officer is Weather Operations primary Point of Contact (POC) for daily operational support, at least one SOF/DUTY OFFICER feedback submission per day (any formal listed type of feedback from previous methods) is desired. Web-based or emailed submissions are preferred. For counter forecast briefings (transient, etc.), at least one feedback sheet (web-based or paper copy) per day is desired.

1.9.7. SOFs or aircrews can also provide feedback during liaison visits. Workload and weather conditions permitting, Weather Forecasters will visit the 53 WRS duty desks not less than one (1) day per week when flying is taking place. The 815 AS will be contacted, at a minimum, on Tuesdays and Thursdays. During these visits, discussions will include weather for the day, weather for the next flying day and the next five (5) days, as well as how missions from the previous day were affected by weather. These visits provide an excellent opportunity for aircrews to provide feedback about briefings and MEF products.

1.10. Mishap Procedures. Both Mission Weather Element and Airfield Services Element have a role when Weather Operations is advised of an emergency or mishap. 557th Weather Wing (WW) and the 26 OWS are also involved. See section 4.11. for Aircraft Accident procedures.

1.11. Utilization of Weather Forecasters. Weather Forecasters are not to be tasked for ready augmented positions or functions and details that will take them away from their Mission Weather Element or Airfield Services Element duties.

1.12. Records Management. BOS/WSO will maintain and dispose of records according to Air Force Records Information Management System (AFRIMS) Records Disposition Schedule (RDS).

Chapter 2

MISSION INFORMATION

2.1. General. BOS/WSO supports numerous units with diversified missions assigned to Keesler AFB. Each unit has unique weather requirements to accomplish their mission.

2.2. Geographic Area of Responsibility. BOS/WSO provides mission-tailored weather support to Keesler AFB and for flying areas used by assigned or attached units to Keesler AFB. **Attachment 2** contains maps of Keesler AFB's local flying area, and the routes used by each flying squadron.

2.3. Airframe-Specific Weather Limitations. Airframe weather limitations are based on 11-series AFIs and AFMANs and change often. For a complete list, see the references listed later in this document, most notably the AFMAN 11-202V3, *Flight Operations*, as well as any Technical Orders (TO) that pertain to each aircraft.

2.4. Non-Flying Customer Weather Limitations. The BOS/WSO gains regular exposure to Keesler's customer leadership through briefings and weather discussions. Increased awareness of mission limitations is necessary for proper tailored weather support. See **Attachment 3** for a list of Keesler AFB Units and the weather that impacts their mission.

Chapter 3

MISSION WEATHER ELEMENT

3.1. General. Mission Weather Element includes those actions directly related to completing each customer's daily mission(s) (e.g., Flight Weather Briefings). The Flying Area Mission Execution Forecast (FAMEF) is the primary tool used by the 403 WG to accomplish their mission. FAMEFs are tailored to individual customer requirements. A Mission Execution Forecast (MEF) can also be a DD Form 175-1 *Flight Weather Briefing* or a non-flying forecast for activities that may be affected by weather and require planning.

3.2. Operational Hours. Most mission services, like flying FAMEFs, are only required during normal duty and/or flying hours. BOS/WSO will have at least one person dedicated to Mission Weather Element during normal duty hours/days if unit manning is adequate. Missions outside these times or in times of inadequate manning will be accomplished by forecasters performing Airfield Service Element duties or, if needed and available, by scheduling additional Weather Forecasters during that period. Forecasting services are available 24 hours a day BOS/WSO/26 OWS team. BOS/WSO has a Weather Forecaster on Monday through Friday (Weather Forecasters will be available on standby Saturday and Sunday excluding the 403d UTA weekend when forecasters will be available from 1000L to 1700L on Saturday and 0700L to 2300L on Sunday. The Mission Weather Element Weather Forecaster can be contacted by phone at DSN 597-3305 or COMM 228-377-3305 or by email at **81osf.KeeslerWeather@us.af.mil**. **Note:** The 26 OWS operates continuously. During non-duty hours, the 26 OWS can be contacted at DSN 331-2651 or Commercial 318-529-2651 for forecasting/briefing assistance.

3.3. Flying Area Mission Execution Forecasts. FAMEFs are essentially mission-specific forecasts that are developed using an 8-step process, outlined in DAFMAN 15-129, and will be provided as a web-based briefing, a verbal, or on a DD Form 175-1, *Flight Weather Briefing*. During this process, BOS/WSO will fuse, and tailor products created by strategic and theater weather centers, as well as information supplied by local units (e.g., flying schedule) and agencies. The result is a product/information designed to provide timely, accurate, and relevant weather intelligence to various customers by whatever means proves most effective. FAMEFs must be horizontally consistent with (but not necessarily mirror) products issued by the 26 OWS and AFW-WEBS. However, during rapidly changing conditions, emergencies or when conditions threaten resource protection, BOS/WSO will amend the FAMEF to accurately reflect conditions and back brief the 26 OWS when time permits. The FAMEF will be posted to the 81OSF-Weather SharePoint site for local Keesler aircrews to use. If the primary method is down, verbal briefings and or a DD Form 175-1, *Flight Weather Briefing,* will be used. Also, if Weather Operations loses Internet capability and are unable to complete FAMEFs, they will turn to the 26 OWS for backup support until Internet capabilities return. FAMEFs will be posted twice a day Monday - Friday, three (3) hours prior to the first local area flight for both the morning and afternoon when local area missions are scheduled by Keesler aircrews. The morning FAMEF will be valid one hour prior to takeoff for nine hours. The afternoon FAMEF will be valid from one hour (1) prior to takeoff until local flying is completed. Once local flying is completed or the FAMEF is no longer valid, the blank master FAMEF stating when the next FAMEF will be issued is to be posted. If the afternoon and evening flights are scheduled to land within two (2) hours of the ending valid time of the morning FAMEF, it may be reviewed and extended to cover those flights instead of issuing a completely new product at 1500L. Local Missions are defined as taking off and landing at

Keesler AFB, MS (KBIX), with stops at Gulfport, MS (KGPT), Mobile, AL (KBFM), Stennis, MS (KHSA), Camp Shelby Auxiliary Field (KSH1), Jackson, MS (KJAN), Baton Rouge Municipal Airport (KBTR), with Monroe Regional (KMLU), Ellington Field (KEFD), and Navy New Orleans (KNBG), being alternates. The alternate section will be filled out when any airfield in the local section is forecasted to have ceilings/vis less than 2,000'/3miles, or if requested by the customer. Any other flights must call (228-377-3305) to receive a DD Form 175-1 or verbal weather briefing. Backup dissemination for the FAMEF will be by email, phone or hand delivery. See **Attachment 4** for an example of Tab 1 (FAMEF), Tab 2 (Drop zone/Assault Landing forecasts), Tab 3 (SFMR forecast) and Tab 4 (Steinhawk DZ Winds) SMFR Forecast. Note: Times the FAMEFs are issued are subject to change based on mission requirements.

3.3.1. Amendment Criteria. Forecasters will amend the FAMEF whenever any of the Mission Limiting Thresholds (listed in **Table 2.1.; Table 2.2.; Table 2.3)** are met and not correctly forecasted for. Before amending, the Weather Forecaster will call the flying squadron SOFs/Duty Officers and give them a brief description of why we are amending the FAMEF. The forecaster will then amend the FAMEF and repost it to the 81OSF SharePoint site. The forecaster will then call the SOFs/Duty Officers once again and inform them that the amended FAMEF is posted.

3.3.2. Updates. Weather Forecasters will monitor the FAMEF continuously and update it as required. Updates to the FAMEF, including category changes (representative amendments are not required), may precede the issuance of the Keesler AFB TAF but must be verbally coordinated between Weather Operations prior to any FAMEF changes.

3.3.3. Follow-Up Support. Aircrews are welcome to contact BOS/WSO (228-377-3305) for any post-mission information and/or follow-up support. Feedback information is available in **paragraph 1.9**.

3.3.4. The FAMEF sections are filled out as follows:

3.3.4.1. The Local Flying Area Forecasts and Alternates sections will be filled out in the same manner as a DD Form 175-1.

3.3.4.2. FL Level Winds are Above Ground Level (AGL). Wind direction and speed will be predominate directions with worst case wind speed. If there is an expected shift in direction (IE frontal passage) additional wind directions and speeds will be provided in the block.

3.3.4.3. Hazards. If the box for a hazard is green, that hazard is not forecasted during the forecast period. If the box for a hazard is red, that hazard is forecasted during the forecast period. Aircrews may contact Weather Operations for the information or other DOD approved sources.

3.3.4.4. Today's Light Data is provided in local time and is based on the information for Keesler AFB.

3.4. Flight Weather Briefings. Weather Forecasters will provide verbal or traditional Flight Weather Briefings (DD Form 175-1) to aircrews as requested and following BOS/WSO duty priorities listed in **Table 1.1**. Briefings for Keesler Air Force Base (KEESLER AFB) aircrews departing from KEESLER AFB will be provided by BOS/WSO between 0600L and 2300L Monday –Friday excluding federal holidays and approved airfield closures. Briefings will be either at BOS/WSO, building 0233, email or phone. Weather Forecasters require a 2-hour notice prior to briefing time to allow for the most accurate weather data. Aircrews can contact BOS/WSO at 228-377-3305 to request a briefing. Out-of-station mass briefings for special missions require 48-hours advanced notice, are subject to staffing availability, and must be coordinated with BOS/WSO leadership at 228-377-3938. Transient aircrews departing Keesler AFB will receive flight weather briefings from the Keesler BOS/WSO during published duty hours, their home Base Weather Station, or the 26 OWS. EXCEPTION: All strategic mobility missions will be passed to an IFM (Integrated Flight Manager) at 618 AOC at Scott Air Force Base). The 26 OWS can be contacted by phone (DSN 331-2651) or via web access **https://26ows.us.af.mil**, in Operations, click on Request a Briefing. The transient aircrew weather briefing area located in BOS/WSO will be used to obtain briefings from the home Base Weather Station or 26 OWS. These briefings are normally done face-to-face or sent via email. If email is inoperable, Weather Forecasters can hand deliver the product if manning allows. Also, if BOS/WSO loses Internet capability and are unable to complete FAMEFs, they will turn to the 26 OWS for backup support until Internet capabilities return.

3.4.1. Updates. Weather Forecasters will monitor the DD Form 175-1 until support is no longer required. If updates are needed, forecasters will provide them to the aircrews before leaving Airfield Management. If the aircrew has left Airfield Management, forecasters will contact Keesler AFB ATC and request they contact the aircrew and have the aircrew call BOS/WSO on the PMSV. See **Attachment 5** for an example of the DD Form 175-1, *Flight Weather Briefing*.

3.4.2. If operating off station, WSO is still the primary weather support agency during KEESLER AFB airfield operating hours and that during non-duty hours the 26 OWS takes over that responsibility. WSO will also make sure that if they or the 26 OWS is not available that the DSNs and commercial phone numbers for other Operational Weather Squadrons (OSW)s are made available.

3.5. 815 AS Drop Zone and Camp Shelby Assault Landing Strip Mission Execution Forecast. 815 AS Drop Zone and Assault Landing Strip Mission Execution Forecasts are mission-specific forecasts developed using an 8-step process outlined by DAFMAN 15-129. During the process, BOS/WSO will fuse, and tailor products created by strategic and theater weather centers, as well as information supplied by local units (e.g., flying schedule) and agencies. The end result is a product or information that is designed to provide timely, accurate, and relevant weather intelligence to various customers by whatever means proves most effective. The Drop Zone MEF will be posted on Monday, Tuesday, and Thursday. The Assault Landing Strip MEF will be posted Wednesdays on the same section as the FAMEF. Both will be posted on the81 OSF-Weather SharePoint site. If the 815 AS needs the Drop Zone MEF at any other time, they will coordinate with BOS/WSO on when the product is needed. If BOS/WSO is not able to post the briefing, it can be hand delivered by a forecaster if manning allows, or the 815 AS can send a runner for pickup at BOS/WSO. See **Attachment 4** for an example of the 815 AS Drop Zone Mission Execution Forecast.

3.5.1. Amendment Criteria. Weather Forecasters will amend the Drop Zone or Assault Landing Strip MEF whenever any of the Mission Limiting Thresholds (listed in **Table 2.4**) are met and not correctly forecasted for. The Weather Forecaster will call the 815 AS Duty Officer and give them a description of why we are amending the Drop Zone MEF.

3.5.2. The 815 AS Drop Zone MEF consists of two pages: first page includes the daily FAMEF, and the second page includes forecasts for the BOFA and Ralffie Drop Zones and the Assault Landing Strip (KSH1). In the event of a change in times, locations, or delivery method, 815 AS leadership will contact BOS/WSO.

3.5.2.1. The first page is the normal daily FAMEF. All times and locations have been coordinated with 815 AS leadership.

3.5.2.2. The second page is a local document created in coordination with 815 AS leadership and will be filled out in the following manner:

3.5.2.2.1. FL Winds/Temp: Altitudes are in AGL. Wind direction and speed will be predominate directions with worst case wind speed. If there is an expected shift in direction (IE frontal passage) additional wind directions and speeds will be provided in the block.

3.5.2.2.2. Forecasted surface winds, temperature, cloud cover, visibility/weather and altimeter will be in the same manner as that data would be filled out on a normal DD Form 175-1.

3.6. Missionwatch. This term is used to describe the process by which BOS/WSO monitors the weather for all missions. All on-site meteorological and commercial data sources (METSAT, Weather Radar, Websites, etc.) may be used to accomplish this task and is the primary responsibility of the Mission Weather Element. It is through this method that MEF amendments/updates are accomplished. During rapidly changing weather, BOS/WSO will amend/update the MEF as required and contact the owning SOF/Duty Officer to pass on critical changes. Significant changes include hazards not briefed, destination and alternates all reflecting mission limiting threshold conditions not previously briefed, any PIREPs or Sigmets received along the route which contain hazards not briefed on departure, etc.

3.7. Pilot-to-Metro Service. Weather information is available during normal duty hours on frequency 267.4 MHz to provide aircrews with operational weather support. For aircraft outside the range of our PMSV system, when the station is closed, or during outages, they will have to obtain PMSV support from another agency. For phone patches to the 26 OWS, aircrews can contact DSN 331-2651 or COMM 318-529-2651.

3.7.1. BOS/WSO will disseminate PIREPs locally when they reflect operationally significant weather within 50NM of Keesler AFB (reports of any icing, turbulence, low-level wind shear, thunderstorm activity, tops/bases of clouds, flight visibility, and obstructions to vision). PIREPs will be disseminated long-line IAW AFMAN 15-124, *Meteorological Codes*.

Chapter 4

AIRFIELD SERVICES ELEMENT

4.1. General. Airfield Services Element includes those actions that affect the Keesler aerodrome or the base as a whole. Examples include weather observations, TAFS, and resource protection. Resource protection, however, will be addressed in **Chapter 7**.

4.2. Operational Hours. BOS/WSO Airfield Services Element is available 24 hours a day Monday through Friday. Weather Forecasters will also be available from 1000L to 1700L on Saturday and 0700L to 2300L during the 403 WG UTA weekend. 403 WG will coordinate with 81 OSF leadership and BOS/WSO LF for the times they need support. There is no Airfield Services Element scheduled on federal holidays or approved airfield closures. Airfield Services Element Weather Forecasters can be contacted by phone at DSN 597-4397 or COMM 228-377- 4397 or by email at **81OSF.KeeslerWeather@us.af.mil**. When the airfield is closed, a member of BOS/WSO is on standby to be recalled in the event the airfield is opened outside of normal operating hours.

4.2.1. There will be a BOS/WSO standby Weather Forecaster available at all times who will man BOS/WSO outside the “normal” operating hours mentioned in **paragraph 4.2**. This Weather Forecaster will man BOS/WSO when severe watch or warning criteria is expected or present (excluding lightning watches/warnings) or for aircraft arrivals or departures. Standby forecasters will be recalled by the Keesler Command Post (recall will be initiated due to notification by 26 OWS or opening of the Keesler AFB airfield). Once notified of the recall, the standby forecaster will arrive as soon as possible after being recalled. Standby forecasters will also be available for mission requirements, of which a remote briefing from the 26 OWS will not suffice.

4.3. Terminal Aerodrome Forecast. Keesler AFB WSO will issue the TAF every eight hours at 0500 Central Time (CT), 1300 CT, and 2100 CT (TAFs will not be issued on airfield closure days); The TAF issued prior to airfield closure will contain the remark “LAST NO AMDS AFT ddhh NEXT ddhh" (dd is the date and hh is the time when the airfield closes, the next dd is the date and hh is the time when the new TAF is required). When the WSO unit has been closed, the next initial TAF is issued not more than three (3) hours prior to the airfield opening. The TAF covers a 30-hour period from the time of issuance and is disseminated by BOS/WSO via the ADS. TAFs apply to the area within a 5NM radius of a designated center point of KEESLER AFB. TAF specification and amendment criteria are listed in **Attachment 9** and **Attachment 10** contains examples of typical KEESLER AFB TAFs.

4.3.1. If the primary means of dissemination is inoperable BOS/WSO will relay the TAF to the Tower, Airfield Management, and, Keesler CP. It will be entered into the Automated Weather Network (AWN) by the Keesler BOS/WSO, another Base Weather Station, or via an approved Internet Site (e.g., the Air Force Weather Web Services [AFW-WEBS]).

4.4. Observations. Observations are taken, recorded, and disseminated IAW AFMAN 15-111, *Surface Weather Observations*. The AF Form 380D3**,** *Federal Meteorological Surface Weather Observations (METAR/SPECI)***,** will be used as the form/method for observation record. Observations typically contain the following elements: observation type; date/time; wind speed and direction (true); prevailing visibility; runway visual range (RVR) as required; present weather; cloud groups; temperature/dewpoint; altimeter setting (ALSTG); and additional remarks. There are two types of observations (described in **paragraph 4.4.3** and **paragraph 4.4.4)** that are seen by customers. **Attachment 10** contains examples of the different types of observations.

4.4.1. AN/FMQ-19 AMOS. The AN/FMQ-19 AMOS is the primary instrument used to take observations at Keesler AFB. During the hours that the airfield is closed, the AN/FMQ-19 AMOS will be in Automated Mode. During this time, the AN/FMQ-19 AMOS will send observations automatically. During the hours that the airfield is open (0800L-2300L Mon-Fri, and 1100L-1700L), the system will be supplemented or backed up, when necessary, according to **paragraph 4.4.1.1** and **paragraph 4.4.1.2**. **Note:** The AN/FMQ-19 AMOS will also be augmented if a forecaster is called in for resource protection when the airfield is closed.

4.4.1.1. Supplementing the AN/FMQ-19 AMOS. Supplementing is the method of manually adding meteorological information to an automated observation generated by the AN/FMQ-19 AMOS that is beyond its capabilities to detect or report. Supplementation is mandatory when any of the following are met (see **Table 4.1**.). Supplementing is also mandatory when Local authorities (wing/garrison or senior mission commander) deem an operationally significant parameter, beyond the capability of the system to measure requires supplementation based on their operational risk assessment (e.g. if the airfield remains open when visibility falls below 1/4 mile, then visibility values below 1/4 mile should be supplemented).

Table 4.1. Supplementary Weather Conditions.

Mandatory Supplementary Weather Conditions - Body of Report (Note 1.)
Tornado (+FC) (Note 2) (Note 3)
Funnel Cloud (FC) (Note 2) (Note 3)
Waterspout (+FC) (Note 2) (Note 3)
Hail (GR) (local warning criteria only)
Volcanic Ash (VA)
Sandstorms (SS) or Dust storms (DS) (Note 4) (If local warning required)
Ice Pellets (PL)
Visibility $\leq$ 1/2 mile (800 meters) and or ceilings $\leq$ 200’
Mandatory Supplementary Weather Conditions- Remarks Section of Report (Note 1.)
Funnel Cloud (Tornadic Activity _B/ E(hh)mm_LOC/DIR_(MOV)) (Note 2)
Snow Depth (Note 4) (only during airfield operating hours)
Tower Visibility (Include a tower visibility remark in the next METAR or SPECI when either the surface prevailing visibility or the tower visibility is less than 4 statute miles and differs from the surface prevailing visibility by a reportable value.)
Notes: 1. References for coding of augmentable weather conditions are in AFMAN 15-111, Chapter 13. 2. The immediate reporting of funnel clouds takes precedence over any other phenomena. 3. Log on to AMOS or ADS and be prepared to supplement for tornadic activity whenever a warning is issued, or a watch is valid for the phenomena. 4. All Remarks and Additive Data references are provided in AFMAN 15-111, Attachment 3.

4.4.1.2. Backing up the AN/FMQ-19 AMOS. Backup (editing) is the process of providing meteorological data, documentation, and/or communication of an AN/FMQ-19 AMOS weather observation when the primary automated method is unavailable or unrepresentative. Backup of the AN/FMQ-19 AMOS by the Weather Forecaster provides the minimum acceptable level of available data for meteorological operations and the support of aviation operations in the event of a partial or total AN/FMQ-19 AMOS failure.

4.4.1.2.1. The following functions of the AN/FMQ-19 AMOS will be backed up if the Weather Forecaster judges the difference between what the AN/FMQ-19 AMOS is reporting and manually observed conditions are operationally significant: wind speed and direction, visibility, selected present weather elements and obscurations, sky cover up to and including 12,000 feet, temperature and dew point temperature, altimeter setting, some remarks and additive data. In these instances, Weather Forecasters must log out the AN/FMQ-19 AMOS with maintenance to have the sensor evaluated and the issue corrected.

4.4.2. Official Observing Site.

4.4.2.1. Official Automated Observing Site. This is the location of the primary sensor AN/FMQ-19 AMOS. At Keesler AFB, this location is just west of the 22 end of the runway, across from the Marina.

4.4.2.2. Official Manual Observing Site. The physical location from which the Weather Forecaster visually determines sky condition, present weather and prevailing visibility is the official observing site. Keesler's official Manual observation point is on the flight-line side of bldg 0233 and approximately 20 yards in front of Airfield Management's flight-line entrance door. This observation can be supplemented with conditions observed from the stairwell next to BOS/WSO (weather permitting) and the southeast external corner of bldg 0233. This is the location from where Weather Forecasters will back-up or supplement the AN/FMQ-19 AMOS.

4.4.3. METAR (Routine Meteorological Observation Report). METAR observations are taken and disseminated every hour between 55 and 59 minutes after the hour. METAR observations are disseminated both locally and long-line.

4.4.4. SPECI (Special Observation). SPECI observations are taken whenever certain weather events, defined in AFMAN 15-111 and listed in **Attachment 11**, are observed at KEESLER AFB. These observations are disseminated both locally and long-line.

4.5. Basic Weather Watch (BWW). A BWW is a process by which Weather Forecasters periodically recheck weather at intervals not to exceed 20 minutes. Because Keesler AFB is an automated station, a BWW is performed during AMOS augmentation.

4.6. Continuous Weather Watch (CWW). A CWW is a process by which Weather Forecasters continuously watch weather conditions without performing any other duties. There is no established requirement for a continuous weather watch at KEESLER AFB.

4.7. Cooperative Weather Watch. This is the process by which Air Traffic Control (ATC) personnel, SOF/DUTY OFFICERs, flying units and Security Forces (SF) personnel report changes in weather conditions to the duty Weather Forecaster (228-377-3305/4397). Of primary concern is the occurrence of previously unreported weather conditions that could affect flight safety, or which could be critical to the safety or efficiency of other local operations and resources. These changes may be included in the next scheduled surface observation, become the basis for an entirely new observation or be included as reports of differing conditions in the remarks section of an observation in progress. Cooperative weather watch inherently dictates close contact between the KEESLER AFB BOS/WSO and the previously mentioned organizations and/or personnel, especially ATC. BOS/WSO leadership will be responsible for ensuring cooperative weather watch compliance by coordinating with associated cooperative weather watch unit leadership. Cooperative Weather Watch techniques are included in the ATC Indoctrination training that's provided by Weather to new Tower personnel. After a basic weather test has been passed, the Tower personnel's training is then recorded in their training records.

4.7.1. ATC personnel will notify BOS/WSO when / if:

4.7.1.1. A PIREP is received. ATC personnel will pass airborne weather information and PIREPs to BOS/WSO IAW FAAO 7110.65, *Air Traffic Control*.

4.7.1.2. Anytime changes in clouds are noticed (e.g., the ceiling raises or lowers or cloud cover changes from scattered to broken/overcast or vice versa).

4.7.1.3. Tower visibility decreases to less than, or if below, increases to equal or exceeds 4, 3, 2, 1 3/4, 1 1/2, 1 1/4, 1, 3/4 or 1/2 miles.

4.7.1.4. Precipitation begins, ends or changes in form (e.g., rain becomes ice pellets, turns into snow, etc.).

4.7.1.5. Thunder is heard.

4.7.1.6. A tornado, funnel cloud or waterspout is observed and/or disappears from sight.

4.7.1.7. Lightning is observed at any distance from the control tower.

4.7.1.8. When prevailing visibility is one mile or less or the runway visual range (RVR) is 6,000 feet or less, report changes to the high intensity runway light setting to BOS/WSO.

4.7.1.9. When ATC judges the automated observation to be unrepresentative of actual conditions. At that time, the weather forecaster will evaluate the situation to determine if augmentation is necessary.

4.7.1.10. IAW FAAO 7110.65, Air Traffic Control Tower personnel must use the lower prevailing visibility from either the Weather Forecaster (surface) or from the air traffic controller (tower level) for aircraft operations when the visibility at either location is less than four miles.

4.7.2. Security Forces personnel will:

4.7.2.1. Report observed damaging or threatening weather conditions to BOS/WSO or to the Keesler Command Post (CP) for relay to BOS/WSO. Early reports of undetected phenomena could be critical for warning other personnel who require shelter.

4.7.2.2. Collect area impacts of severe weather events and report them to BOS/WSO or the CP for relay to BOS/WSO. **Note:** Threatening weather includes close proximity lightning, funnel clouds, tornadoes, waterspouts, hail of any size, wind gusts affecting unsecured objects, etc.

4.7.3. Fire Department personnel will:

4.7.3.1. Provide CBRNE support to base personnel.

4.7.3.2. Collect area impacts of severe weather events and report them to BOS/WSO or the CP for relay to BOS/WSO.

4.7.4. Locally stationed aircrew will report PIREPs IAW Flight Information Handbook Section C and 11 series AFIs. They will:

4.7.4.1. Report unusual or unforecasted weather to appropriate control agency without delay.

4.7.4.2. Provide PIREPs over PMSV to include weather conditions experienced during climb and descent.

4.7.4.3. Provide PIREPs for the following criteria, which will be disseminated as “Urgent Upper Air” reports by weather personnel.

4.7.4.3.1. Tornado/waterspout or funnel cloud.

4.7.4.3.2. Thunderstorms **Note:** The requirement for thunderstorms refers to the occurrence of an area of widespread activity, thunderstorms along a line with little or no space between individual storms, or thunderstorms embedded in cloud layers or concealed by haze. It does not refer to isolated or scattered thunderstorms not embedded in clouds or not concealed by haze.

4.7.4.3.3. Tropical cyclones.

4.7.4.3.4. Squall line.

4.7.4.3.5. Severe icing.

4.7.4.3.6. Severe or extreme turbulence, including Clear Air Turbulence (CAT).

4.7.4.3.7. Mountain wave turbulence.

4.7.4.3.8. Widespread dust storm or sandstorm.

4.7.4.3.9. Low-level Wind Shear (LLWS) when the fluctuation in airspeed is 10 knots or more.

4.7.4.3.10. Hail.

4.7.4.3.11. Volcanic eruption and/or ash when reported by any source, in the air or on the ground.

4.7.5. Local SOF/DUTY OFFICERs will:

4.7.5.1. Relay all PIREPs and summary of flight weather conditions experienced by their aircrew to the Weather Forecasters as received.

4.7.5.2. Provide feedback in accordance with **Paragraph 1.9** of this instruction.

4.7.6. Weather Forecasters will:

4.7.6.1. Notify the Air Traffic Control Tower as soon as possible, whenever the prevailing visibility at the weather unit's observation point decreases to less than or increases to equal or exceed four (4) miles (6,000 meters).

4.7.6.2. Re-evaluate surface prevailing or sector visibility as soon as practicable, upon initial receipt of a differing Air Traffic Control Tower value, and upon receipt of subsequent reportable changes at the Air Traffic Control Tower level.

4.7.6.3. Use Air Traffic Control Tower values of prevailing (during back up operations) or sector visibility (during supplemented operations) as a guide in determining the surface visibility when the view of portions of the horizon is obstructed by buildings, aircraft, etc. The presence of a surface-based obscuration, uniformly distributed to heights above the level of the Air Traffic Control Tower, is sufficient reason to consider the weather unit's prevailing visibility to be the same as the Air Traffic Control Tower level.

4.7.6.4. Contact the Air Traffic Control Tower if any of the following is expected to occur at Keesler AFB within the next hour: (This provides Air Traffic Control personnel advance notice to monitor for changing conditions).

4.7.6.4.1. Tornadic Activity.

4.7.6.4.2. Ceilings below 2,000 feet.

4.7.6.4.3. Visibility below 4SM.

4.7.6.4.4. Thunderstorms.

4.7.6.4.5. Lightning.

4.7.6.4.6. Precipitation.

4.7.6.4.7. Winds $\geq$ 50kts.

4.7.6.5. Weather Forecasters will reevaluate the weather conditions whenever reliable source (i.e., ATC, pilots, local law enforcement, etc.) reports weather conditions different from the last disseminated observation (e.g., different ceiling height, visibility, present weather). Based on reevaluation of the different weather conditions reported and local policy, Weather Forecasters will:

4.7.6.5.1. Generate a SPECI observation (during supplemented/back up operations) if the different conditions warrant immediate dissemination.

4.7.6.5.2. Include the differing conditions in the next required METAR or SPECI observation (during supplemented/back up operations) if the conditions alone do not warrant immediate dissemination.

4.7.6.5.3. Incorporate into any Mission Execution Forecast that is valid.

4.7.6.6. Issue watches warnings and advisories to base agencies.

4.7.6.7. BOS/WSO will disseminate PIREPs locally when they reflect operationally significant weather within 50NM of Keesler AFB (reports of any icing, turbulence, low-level wind shear, thunderstorm activity, tops/bases of clouds, flight visibility, and obstructions to vision). PIREPs will be disseminated long-line IAW AFMAN 15-124, *Meteorological Codes*.

4.7.6.8. Contact the ATC Tower and let them know that our PMSV is out. Ask them to inform aircrews trying to contact Keesler Metro that our PMSV is out, and they will need to contact another location for assistance.

4.7.6.9. Provide the Fire Department, Bio-Environmental, and Emergency Management with Chemical, Biological, Radiological, Nuclear, and High Yield Explosive (CBRNE) inputs using the Chemical Down Wind (CDM) and Effective Downwind (EDM) Messages and wind data as requested.

4.7.6.10. Contact the 26 OWS and pass on pertinent information as necessary.

4.8. METWATCH. METWATCH provides an organized approach for weather personnel to maintain situational awareness of the current/future meteorological situation within a designated area(s). This process involves notifying supported units and updating any forecast products when pre-established weather conditions or unforecasted changes (timing, location or forecast values) in weather occur or are expected to occur. All on-site meteorological and commercial data sources (METSAT [Meteorological Satellite], weather radar, websites, National Oceanic and Atmospheric Administration [NOAA] Weather Radio, etc.) may be used to accomplish this task. The 26 OWS will perform a continuous Terminal METWATCH for KEESLER AFB with Weather Forecasters acting as the “eyes forward” for the 26 OWS by providing immediate feedback on current or short-

term anticipated changes in weather conditions. The 26 OWS will also perform Flight and Route METWATCH for transient flights they briefed, which are departing from KEESLER AFB. BOS/WSO will perform Flight and Route METWATCH for transient flights departing KEESLER AFB that were briefed by Weather Forecasters. **Note:** Though similar to MISSIONWATCH (**Paragraph 3.6.**), transient missions (cross country, etc.) are not considered "local," though the mission's origins may have started/ended at KEESLER AFB.

4.8.1. At a minimum, METWATCH procedures will consist of the following:

4.8.1.1. Weather Forecasters will monitor conditions at KBIX, KGPT, KBFM, KJAN, and KBTR; 26 OWS and SPC hazard charts; Space Weather forecasts; SIGMETs/PIREPs and Lightning data from various sources (e.g., AFW-WEBS, FMQ-19).

4.8.1.2. Weather Forecasters will METWATCH for Forecast specification and amendment criteria as detailed in **Attachment 9** of this instruction; watch, warning and advisory criteria as outlined in **Chapter 7** of this instruction; and hurricane/tropical storm advisory criteria.

4.8.1.3. When Weather Forecasters are supplementing or backing up the AN/FMQ-19 AMOS, they will recheck weather conditions, at intervals not to exceed **20 minutes** since the last observation/recheck, when any of the criteria below are observed or forecast to occur within one (1) hour:

4.8.1.3.1. Ceiling decreases to less than or, if below, increases to equal or exceed 1,500 feet. Use PIREPs to aid in determining ceiling heights.

4.8.1.3.2. Prevailing Visibility (be it from Tower or Weather) decreases to less than, or if below, increases to equal or exceed 3SM.

4.8.1.3.3. Precipitation (any form).

4.8.1.3.4. Fog or Mist.

4.8.1.3.5. Thunderstorm or lightning activity within 5NM of Keesler AFB.

4.8.1.3.6. Conditions that require a SPECI.

4.8.1.4. When forecast conditions change during the METWATCH, Weather Forecasters will notify BOS/WSO Mission Weather Element Weather Forecaster.

4.9. Eyes Forward Process/Procedure. This process/procedure is BOS/WSO role in allowing for the integration of weather data, meteorological satellite imagery, lightning detection readouts and non-standard weather data systems to create an integrated weather picture and near-term forecast for the 26 OWS. BOS/WSO provides meaningful meteorological information not contained in the coded observations to the 26 OWS as an integral part of the METWATCH process (phone calls, emails, etc.). Additionally, BOS/WSO integrates the current state of the atmosphere into an understanding of future impacts on the forecast conditions and communicates these impacts to the 26 OWS, as needed.

4.10. Wet Bulb Globe Temperature Index (WBGTI). WBGTI is a predictor of physiological strain in hot environments. It takes into account the effects of humidity, air speed, air temperature and the radiant heating factor (from the sun). The WBGTI is not the same as the Heat Index. Weather Operation will support Thermal Injury Prevention Program (TIPP) by providing current weather information through the online JET program. The BOS/WSO will also assist the TIPP with obtaining and interpreting historical weather data and information, to include the local installation future deployed locations, and other operational needs.

4.10.1. The 81 TRW Bio-Environmental Flight will calculate and disseminate the WBGTI to the Blake Fitness Center and the Keesler CP IAW DAFI 48-151 Thermal Injury Prevention Program. This information is disseminated from 0800L to 1630L Monday through Fridays. The Blake Fitness Center will provide back-up calculation and dissemination on weekends, NTD Fridays and holidays. The WBGTI reading can be obtained from the KEESLER AFB Homepage. If WBGTI is above the green flag status at 1630L, WBGTI will continue to be monitored until a green flag status is reached.

4.11. Aircraft Mishaps. In the event that BOS/WSO is notified of an aircraft mishap, the following procedures will be followed:

4.11.1. Complete all actions of Tab A and/or Tab B to Appendix 2 of 81 TRW IEMP 10-2 in the Emergency Actions Binder.

4.11.2. Begin an Events Status Log, found in the Emergency Actions Binder.

4.11.3. Determine the call sign of the downed aircraft from Airfield Management or the Command Post.

4.11.4. Conduct Gibson Ridge data save for one hour before and one hour after the mishap.

4.11.5. Print out and save electronically the most current satellite (WV, Visual and IR) picture.

4.11.6. Make copies of all products, numerical, graphical, etc. used to create the flight weather briefing, to include any PMSV contact between METRO and the mishap aircrew.

4.11.7. Collect the TAFs and observations for departure point, destination and any alternate(s). Collect any information from five (5) hours before to one (1) hour after the mishap occurred. Upon notification or sighting, at or near the station, of an Aircraft Mishap, the Weather Forecaster will take and record a full element SPECI observation (print a copy of the 3803 and write the remark “ACFT MISHAP” in Col 13, but don’t transmit the remark.

4.11.8. Create a CD with the data collected electronically.

4.11.9. Collect the upper air soundings along the route and nearest the mishap site. Print a copy of each of these locations.

4.11.10. Make a copy of all watches, warnings, and advisories for KBIX.

4.11.11. Only notify 557th WW team chief (DSN 271-2586) of the Mishap/Accident to request they initiate a data save for data that we cannot find or save.

4.11.12. Notify 26 OWS (DSN: *331-2622*) to initiate data-save procedures for any OWS products we use to prepare the flight weather briefing for any weather related or ground mishaps requiring Operational Report 3 (OPREP-3).

4.11.13. Make a copy of all data that will be given to the Safety Office. (This is done to ensure we have a copy of all materials). Give this to the Lead Forecaster.

4.11.14. Place all data in an envelope and label it with the aircraft call sign, date, office symbol, and office phone number. A member of the 81 OSF staff will deliver the envelope to the Safety Office and be prepared to brief the Safety Investigation Board on daily weather conditions expected during the investigation. **Note:** Do not release the names of personnel involved in an aircraft mishap. Refer all inquiries from non-mishap response personnel to Public Affairs.

4.12. Weather Sensors. Upon notification from ATC of a completed runway change, BOS/WSO will change the appropriate weather sensors to match the active runway.

4.13. Weather Equipment. BOS/WSO has sensors to measure cloud base heights, wind direction and speed, atmospheric pressure, rainfall, Runway Visual Range, visibility atmospheric phenomenon, dew point and ambient air temperature. Weather equipment will be discussed further in **Chapter 9.**

4.14. Keesler AFB five (5)-Day Forecast. Every day that BOS/WSO is open, forecasters will produce a five (5) Day forecast presentation for Keesler AFB by 0730L Mon-Fri and 1100L Sat-Sun. This product is created for non-operational use only and is not amended. This product is posted on the 81 OSF-Weather SharePoint site. The five (5) Day forecast is created as a Power Point presentation and includes slides for: weather graphic of today's weather for the U.S., satellite loop of the U.S., radar loop of the U.S., weather graphic for tomorrow's weather for the U.S., five (5) Day forecast for Keesler AFB and the monthly climatology of Keesler AFB. During hurricane season (1 Jun - 30 Nov), a tropical satellite image of the Atlantic Ocean and storm tracks for any tropical systems affecting Keesler AFB will be provided. The five (5) Day Forecast is provided to Keesler AFB units for use to plan their activities over the next five days. If more specific weather forecasts are needed, base agencies can contact BOS/WSO at 228-377-3305/4397. See **Attachment 7** for an example of the Keesler AFB five (5) Day Forecast.

4.15. Hurricane Bulletin. BOS/WSO will use the tropical cyclone forecast issued by the National Hurricane Center (NHC). BOS/WSO will then use the HURREVAC program graphics portion of the Keesler AFB Hurricane Bulletin. No deviation from the official forecast position, track, movement, maximum wind speed, or intensity trend is authorized. BOS/WSO will use the Mission Execution Forecast Process (MEFP) to tailor the official tropical cyclone forecasts into a specific mission forecast product for our supported customers. Tailoring may include factors such as specific local effects such as terrain or relative position to the storm. This bulletin is normally sent to base agencies via email and posted on the 81 OSF-Weather SharePoint site. If email is down, BOS/WSO will send the bulletin via runner to the Command Post.

4.15.1. The Hurricane Bulletin is provided for non-operational use only and will only be created during the hours that BOS/WSO is open. The bulletins do not provide any guidance on evacuating timetables or HURCON recommendations. That information will be provided to Wing leadership, and decisions on evacuating or HURCON changes will come from the Wing Commander. The bulletin contains the following sections: title section (storm category, name, bulletin number and the date and time the bulletin is valid for), map and legend (map includes:

current location, forecast track, cone of uncertainty and any watches or warnings valid), current information (position, movement, current category, wind speeds, location from Keesler AFB, closest point of approach (CPA) of Keesler AFB, and any watches or warnings for the Keesler AFB area), storm information (information on the storm, effects on Keesler AFB and landfall forecast) and other information (HURCON and hurricane category descriptions). See **Attachment 6** for an example of the Keesler AFB Hurricane Bulletin.

4.15.2. BOS/WSO will provide the most up-to-date weather briefings regarding any tropical cyclone activity anytime deemed necessary by Keesler Disaster Response Force (DRF). This information will be relayed to the DRF upon request. The briefing location will be determined by the DRF coordinator. Adequate time will be provided for BOS/WSO to evaluate the latest (NHC) advisory (dissemination times vary), compile and prepare other pertinent information like divert base forecasts and provide meteorological input/advice. The NHC advisories provide the "official forecast" information and will not be deviated from when briefing base leadership, etc. These advisories will provide the foundation for BOS/WSO forecast products (wind, sea conditions, etc.). BOS/WSO forecast inputs, however, may provide additional guidance to senior leadership based on their experience and training. Refer to the 81 TRW PLAN 10-2, *Installation Emergency Management Plan (IEMP)* for further guidance.

Chapter 5

STAFF WEATHER ELEMENT

5.1. General. Staff Weather Element includes those briefings provided primarily by BOS/WSO leadership. These briefings are a specialized type of MEF focused on a particular event/audience. Examples include, but are not limited to, staff and DRF meetings.

5.2. Operational Hours. Most staff services are provided during normal duty hours, *0700L to 1600L Monday-Friday*. Contingency, hurricane, and exercise briefings are provided as needed. BOS/WSO leadership can be contacted by phone at DSN 597-3938/0486 or COMM 228- 377-3938/0486 or by email at **81OSF.KeeslerWeather@us.af.mil**.

5.3. Staff Briefings. Staff weather briefings for the 81 TRW will be provided as required. Standard information includes current surface analysis, current satellite loop, current radar loop, 24 hr surface forecast map, 5-day weather outlook and anything else pertinent to mission/season.

5.4. Weather Indoctrination Training. BOS/WSO will provide weather indoctrination training and initial weather certification to ATC personnel assigned to 81 OSF, IAW AFMAN 13- 204V3, *Air Traffic Control*. After ATC personnel complete the training, they will take a short test on the information and receive a familiarization tour of the weather station. ATC and other agencies requiring similar training should contact BOS/WSO leadership at 228-377-3938 to schedule an appointment.

5.5. Instrument Refresher Course (IRC) Briefings. BOS/WSO provides IRC briefings as required by course scheduling in accordance with AFMAN 11-210, *Instrument Refresher Program (IRP),* and DAFMAN 15-129. The weather portion of the briefing consists of an overview of BOS/WSO, Airfield and Mission Services, BOS/WSO capabilities, BOS/WSO and OWS responsibilities, resource protection, seasonal/regional weather, space weather impacts and procedures for obtaining weather support at other locations.

5.6. Supervisor of Flying Briefings. IAW AFI 11-418 and AETC Sup 1, BOS/WSO will present upcoming seasonal weather challenges at quarterly SOF/DUTY OFFICER briefings. This will also be the venue to address SOF/DUTY OFFICER weather support issues.

5.7. Disaster Response Force (DRF) Briefings. BOS/WSO will provide weather briefings as required for DRF meetings. This includes exercises, hurricane (HurrWatch Team support), deployment briefings, etc. Each briefing will be tailored to provide the appropriate weather intelligence required by wing leadership. The 81 TRW PLAN 10-2, IEMP outlines specifics on Emergency/Crisis Action Response as well as Force Protection and Emergency Actions resulting from weather events/natural disasters.

5.7.1. Pre-deployment Planning. Per KEESLER AFB Deployment Plan 10-403, BOS/WSO will brief all available weather information to include forecasts and climatology for the local area, Aerial Port of Embarkation (APOE), Area of Responsibility (AOR) and Aerial Port of Debarkation (APOD)/Final Destination (DEST) at the deployment concept briefing. If there is no concept briefing, BOS/WSO will relay information to the Installation Deployment Officer (IDO) and/or Deployment Control Center (DCC) as requested.

5.8. Climatology Briefings. Climatology information will be provided as requested. Additionally, monthly climatology will be presented at Wing staff meetings.

5.9. Flight Information Publication (FLIP) Weather Updates. BOS/WSO is responsible for ensuring all weather information in the FLIP is accurate. All weather-related updates will be requested through the Airfield Management FLIP Manager (Airfield Manager), 81 OSF/AFM. The FLIP manager will process the information to AFFSA/OL-D. Updates will fall in one of three categories: revisions, changes, or corrections.

Chapter 6

SPACE WEATHER SUPPORT AND SERVICES

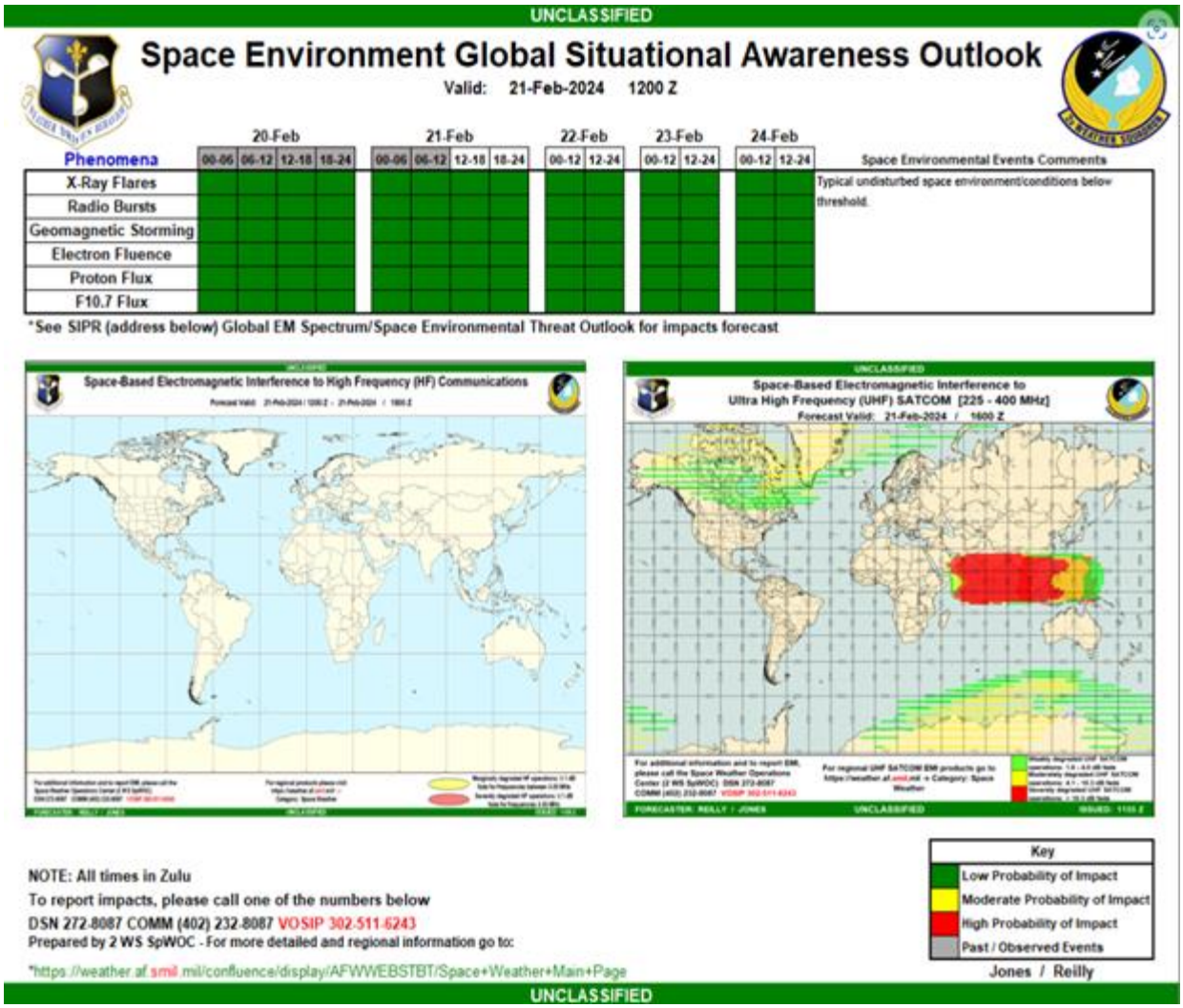
6.1. General. Many KEESLER AFB/USAF communications systems use sky wave and transionospheric propagation (High Frequency [HF], Very High Frequency [VHF], Ultra High Frequency [UHF] and Satellite Communications [SATCOM]); which can be rendered useless by electromagnetic radiation from the sun. This chapter contains some brief information regarding space weather limitations, alerts and warnings and products available to Keesler personnel.

6.2. Limitations. Like terrestrial weather, there are numerous factors that influence space weather. One of the biggest limitations we have in identifying and forecasting space weather is a lack of sensors/data. Additionally, given the speed of solar wind and light, our ability to provide useful lead-times for significant space events is extremely limited.

6.3. Space Weather Alerts and Warnings. With a registered account, broad coverage CONUS-based space weather products (graphics) are available at **https://weather.af.mil/confluence/display/AFWWEBSTBT/Space+Weather+Main+Page** webpages and will be used to update the space weather portion (HF and UHF) of the MEF as well as block #15 (Space Weather) of the DD Form 175-1. GPS Communication will be monitored hourly for specific missions upon request (see **paragraph 6.4**.). Additionally, Space Weather Impacts will be reported through the appropriate link on this site. Bulletins are available through the same link above.

6.3.1. Wx Products: A sample of the daily outlook is provided in **Figure 6.1**. See the links above for current Space Environment Discussions, including potential impacts to Air Force Operations.

Figure 6.1. Sample Space Environment Global Situational Awareness Outlook.



6.4. Products. Most Space Weather products are "now-casts" and/or very short-term forecasts (6-hourly periods), so Mission Weather Element must check websites for updated products (remember to hit refresh on your browser) at least twice throughout their shift and update the FAMEF. For more information concerning these products, please contact the BOS/WSO.

6.4.1. GPS Products. GPS products are primarily "now-casts" and/or <1 hourly forecast. Due to the frequent refresh rate of these products, Mission Weather Element will only need to monitor these products when requested by the customer.

Chapter 7

RESOURCE PROTECTION SERVICES

7.1. General. Certain weather conditions endanger property or life, pose a safety hazard, or adversely impact operations. This chapter contains details on how the BOS/WSO /26 OWS team monitors these phenomena and issues weather watches, warnings and advisories for significant events that are occurring or are expected to occur. Resource protection is accomplished through a joint effort between the 26 OWS and Keesler BOS/WSO. The BOS/WSO is responsible for issuing all forecasted weather advisories, watches, and warnings during normal operational hours or when called in during non-operational hours, the 26 OWS is responsible for this during non-operational hours. BOS/WSO is responsible for issuing all observed warnings and advisories (annotated on the appropriate AF Form 3806, *Weather Watch Advisory Log*, and DAF Form 3807, *Watch/Warning Notification and Verification*). BOS/WSO Conversely, the 26 OWS will issue all observed warnings and advisories during non-operational hours. The goal is to provide the best possible resource protection to KEESLER AFB. If needed, the 26 OWS will be the point for backup support for the Keesler BOS/WSO during severe weather conditions.

7.1.1. Commanders must be familiar with weather watch, warning, and advisory criteria that affect their operations or personnel and ensure that the notification is passed to the lowest level in their unit that may require the notification for personal safety and resource protection.

7.1.2. Weather watches, warnings, and advisories are disseminated by the BOS/WSO or 26 OWS through the ADS. A follow-up call is made to the Keesler CP and ATC tower to ensure receipt. See **Attachment 10** for specific product format and interpretation.

7.2. Weather Watches. A weather watch is a special notice sent to customers indicating that conditions are favorable for the development of a particular type of weather phenomena (e.g., tornadoes, hail, etc.). It provides advanced notification of a potential for weather conditions that could pose a hazard to life or property for which the customer must take protective action. Watches are issued for a 5NM radius of KEESLER AFB.

7.2.1. The BOS/WSO issues all weather watches for KEESLER AFB during operational hours or if present during non-operational hours, the 26 OWS will issue all watches when the weather operations forecaster is not present. Watches normally precede warnings.

7.2.2. All units on KEESLER AFB will, at a minimum, review their protective action procedures when watches are issued.

7.2.3. **Table 7.1.** contains all of the weather watches and desired lead-times issued for KEESLER AFB.

Table 7.1. KEESLERAFB AFB Weather Watch Criteria and Desired Lead-Times.

Criteria	**Desired Lead-Times**	
Tornado	As Potential Warrants	*
Winds ≥ 50 kts and/or Hail ≥ 3/4 Inch Diameter associated with Thunderstorms	As Potential Warrants	*
Winds 35 kts to < 50 kts and/or Hail ½ to < ¾ Inch Diameter, associated with Thunderstorms	As Potential Warrants	*
Surface Winds ≥ 50 kts not associated with Thunderstorms	As Potential Warrants	
Heavy Rain or Snow (≥ 2 Inches in 2 Hours)	As Potential Warrants	*
Heavy Rain (≥ 6 Inches in 12 Hours)	As Potential Warrants	*
Freezing Precipitation	As Potential Warrants	*
Blizzard	As Potential Warrants	*
Lightning Within 5NM of KEESLER AFB	30 Minutes	
Note: A blizzard is defined as a storm with a duration of ≥ 3 hours with sustained winds or gusts ≥ 30 knots, considerable falling and/or blowing snow and prevailing visibility frequently ≤ 1/4SM. All conditions must be met. * Denotes a Weather Watch in which, if issued, requires implementation of the Severe Weather Action Procedures (SWAP) outlined in Chapter 7.7..		

7.3. Weather Warnings. Weather warnings are special notices sent out to customers alerting them that a predefined weather event, of such intensity to pose a threat to life or property (such as lightning, tornadoes, etc.), is expected to occur or is occurring. Warnings are issued for a 5NM radius of KEESLER AFB.

7.3.1. The BOS/WSO issues all weather warnings for KEESLER AFB during operational hours or if present during non-operational hours, the 26 OWS will issue all warnings when the weather operations forecaster is not present.

7.3.2. All units will complete their protective actions upon receipt of weather warnings.

7.3.3. **Table 7.2.** contains all of the weather warnings and desired lead-times issued for Keesler AFB.

Table 7.2. KEESLERAFB Weather Warning Criteria and Desired Lead-Times.

Criteria	Desired Lead-Time	
Tornado	15 Minutes	*
Winds ≥ 50 kts and/or Hail ≥ 3/4 Inch Diameter associated with Thunderstorms	60 Minutes	*
Winds ≥ 35 kts to < 50 kts and/or Hail ≥ ½ to < ¾ Inch Diameter, associated with Thunderstorms	60 Minutes	
Surface Winds ≥ 50 kts not associated with Thunderstorms	60 Minutes	*
Surface Winds ≥ 35 kts but less than 50 kts not associated with Thunderstorms	60 Minutes	
Heavy Rain or Snow (≥ 2 Inches in 2 Hours)	60 Minutes	
Heavy Rain (≥ 6 Inches in 12 Hours)	60 Minutes	
Freezing Precipitation	60Minutes	*
Blizzard	60 Minutes	*
Lightning Within 5NM of KEESLER AFB	Observed	
Note: A blizzard is defined as a storm with a duration of ≥ 3 hours with sustained winds or gusts ≥ 30 knots, considerable falling and/or blowing snow and prevailing visibility frequently ≤ ¼SM. All conditions must be met * Denotes a Weather Warning in which, if issued, requires implementation of the Severe Weather Action Procedures outlined in Paragraph 7.7.		

7.4. Weather Advisories. Weather advisory is a special notice provided to supported customers that alerts them to weather conditions that could affect their operations on KEESLER AFB. Advisories are issued for a 5NM radius of KEESLER AFB. Two types of advisories are issued: observed weather advisories and forecasted weather advisories.

7.4.1. Observed Weather Advisory. Weather advisory issued when the customer does not require advanced notification of a particular weather phenomenon.

7.4.2. Forecast Weather Advisory. Weather advisory issued when the customer requires advance notification of an impending condition with sufficient time to allow protective actions to be taken.

7.4.3. **Table 7.3.** list all weather advisories issued for Keesler AFB. The table includes when the advisory is issued and by what agency. Each agency will provide back-up services to the other when they do not have the capabilities to issue their required advisory.

Table 7.3. Weather Advisory Criteria and Minimum Desired Lead-Times.

Criteria	Desired Lead-Time
Winds ≥ 25 kts to < 35 kts (either BOS/WSO or the 26 OWS Responsibility)	30 minutes
Temp ≤ 32°F for more than 2 hours (either the BOS/WSO or 26 OWS Responsibility)	12 hours
Non-Convective Low Level Wind Shear (Issue only when airfield is open) (either the BOS/WSO or 26 OWS Responsibility)	30 minutes
Temp < 40°F (either the BOS/WSO or 26 OWS Responsibility)	Observed

7.5. Dissemination and Numbering. Watches, warnings, and advisories will be numbered consecutively for an entire month by identifying the type of weather message (watch, warning or advisory) followed by a five-digit number. The first two numbers indicate the current month while the second three numbers indicate the sequence number. For example, the message "Weather Warning 02-005" means the month is February (02), and this is the fifth (005) warning issued in the month. The message "Weather Advisory 12-013" means the month is December (12), and this is the thirteenth (013) issued in the month. See **Attachment 10** for examples. Note: The DAF Form 3806 and/or DAF Form 3807 will be filled out by Airfield Services Element.

7.5.1. Upgrades/Downgrades. Advisories and warnings will be upgraded (i.e., winds increase from 35 knots to 50 knots) or downgraded as required. Upgrades should meet the desired lead-times specified above. Only one forecasted warning, excluding tornado and observed lightning warnings, may be in effect at one time. If a warning is issued for one criterion, and it becomes necessary to warn for another criterion, a new warning, with a new number, will be issued to include all criteria expected. A separate valid time may be specified for each criterion, if necessary.

7.5.2. Extensions or Changes. An extension or change will be issued to change the valid times or duration of a watch, warning or advisory.

7.5.3. Cancellation. Watches and warnings may be cancelled when the weather phenomena are no longer occurring or expected to occur. However, if not cancelled, they will expire at the end of the valid period. Observed advisories will be cancelled when the criteria is no longer occurring and is not expected to occur again in the next hour. The lightning warning and watch will always be cancelled when no longer expected.

7.6. Toxic Corridor. Per DAFI 10-2501, toxic corridor responsibility lies with the Fire Protection Flight with assistance from the Bioenvironmental Flight. BOS/WSO will provide weather data necessary to calculate Toxic Corridors to the Fire Protection Flight and/or the disaster control group, as requested. The 26 OWS may supplement meteorological data to the appropriate Keesler organizations if requested and BOS/WSO is unable to provide the data.

7.7. Severe Weather Action Procedures (SWAP).

7.7.1. SWAP Notification and Response. The purpose of the SWAP is to ensure weather station management (senior, experienced personnel) are notified and respond to both potential and/or actual severe weather events within Keesler's area of responsibility. SWAP activation criteria may include any watch/warning criteria (see **Table 7.1.; Table 7.2** and **Table 7.3**), but will primarily be used for Severe Thunderstorm watches/warnings: Tornadoes/funnel clouds, winds ≥ 50kts, hail ≥ 3/4 inches, heavy rain/snow, freezing precipitation and/or blizzards. During non-duty hours, the 26 OWS along with the CP will contact the Keesler Standby Weather Forecaster when the 26 OWS issues, upgrades, or cancels a watch or warning (does not include lightning watches/warnings or advisories) and recall the Keesler standby Weather Forecaster when a severe weather watch or warning is issued. SWAP activation discretion lies with the on-duty Airfield Services using their experience and guidance from BOS/WSO leadership, including coordination with the 26 OWS (team effort between BOS/WSO and 26 OWS). Further SWAP details are provided in BOS/WSO SWAP SOP. Note: BOS/WSO operational support may be limited to all base units/customers during severe weather events due to limited manning and availability. Limitations to customer support may include Weather Forecasters answer only telephone hotlines, shortened/cancelled weather briefings, long telephone hold periods, slower updates to various FAMEFs, etc.

7.7.2. SWAP Activation Conditions. Once Airfield Services Element expects SWAP activation conditions to occur, he/she will notify the Severe Weather Action Team (SWAT) leader via phone call (see BOS/WSO monthly work schedule for current SWAT leader and contact information, updated as needed, by BOS/WSO Lead Forecaster), follow the SWAP SOPs, and coordinate with the Keesler Weather Forecaster at the 26 OWS. Once contacted, the SWAT leader will determine whether to activate the team. If deemed appropriate, the SWAT leader will report to BOS/WSO as soon as possible (30 minutes ideally) and may call in additional forecasters to assist. The 26 OWS will provide backup SWAT support for BOS/WSO in the event of communication failure, equipment outages, forecaster unavailability, etc. Per DAFMAN 15-129, BOS/WSO will conduct and document a semi-annual exercise. Real world severe weather events can be used in place of these exercises.

7.7.3. Procedures Before and During Potential Severe Weather. Airfield Services Element will keep the SWAT standby person/leader informed of the potential for severe weather. They will immediately notify the SWAT leader whenever the potential for a severe thunderstorm moves within 100NM of KEESLER AFB and/or is expected to affect KEESLER AFB within four (4) hours. The SWAT leader and Airfield Services Element will discuss, analyze, and assess the severe weather threat by reviewing pertinent meteorological information. The goal is to commit earlier, and issue severe weather watches and warnings based more on severe weather indicators and less on actual sightings. If the SWAT leader, and Airfield Services Element determine there is a risk for severe weather, they will implement a contingent of the team, as needed. Otherwise, Airfield Services Element will return to normal operations.

7.7.4. Task Assignments. Once the team is in place (60-minute ideal recall for augmenters), the SWAT leader will assign tasks (monitor weather radar, answer telephones, etc.) and ensure special attention is paid to BOS/WSO Duty Priorities (**Paragraph 1.4.2**.)**.** The members will dedicate their efforts to their assigned tasks and keep the team leader updated on events. Airfield Services Element will maintain primary focus on the changing weather scenario and will coordinate a final decision on which weather products to issue. Mission Weather Element will enhance MISSIONWATCH procedures and contact SOF/Duty Officer's and Duty Desks with any updates necessary.

7.7.5. Response Actions. 81 TRW Severe Weather response actions are outlined in the 81 TRW PLAN 10-2, IEMP.

7.7.6. Post-Event Procedures. Once the SWAT leader and Airfield Services Element have determined the severe weather threat has passed, the SWAT leader will ensure BOS/WSO is returned to "ops normal" (ensure work area is clean, answer/return any pending phone calls, report any severe weather damage [see OPREPs, **Paragraph 7.11**]**,** etc.) before dismissing the SWAT. Any changes to the SWAP procedure will be documented in this instruction (annotated on BOS/WSO master copy prior to the update) and in BOS/WSO SWAP SOP.

7.8. Hurricane/Tropical Storm Bulletin Criteria. Hurricane/tropical storm bulletins are issued whenever a tropical storm or hurricane is within or is forecasted to be within 960NM and has the potential for significant impact on KEESLER AFB, or by direction of 81 TRW or BOS/WSO leadership. The main purpose of the bulletin is to alert base agencies of the potential for winds 50 knots or greater within 96, 72, 48, 24, or 12 hours respectively at KEESLER AFB. These special advisories are based on the National Hurricane Center (NHC) forecasts and outlined in DAFI 10-2501. The NHC website is **http://www.nhc.noaa.gov**. Information will include current storm location, movement, and wind speeds along with forecast positions and wind speeds and will be available at 0400, 1000, 1600 and 2200L, unless there is a change to the forecast. Intermediate special forecasts will be issued whenever there is a hurricane or tropical storm watch, or warning issued. When briefing Wing leadership, there will be no deviations from the official NHC positions, track, movement, and maximum wind speed or intensity trend. Due to the uncertainty of the 72 and 48-hour outlooks, they should be used for planning purposes only (96 and 120-hour forecast positions may also be provided). NHC advisories (official forecast information) are the core of BOS/WSO input to the DRF (**Paragraph 4.15.2.**). NHC data, however, may be used to tailor Keesler forecast products (FAMEFs) with special consideration being given to local terrain effects (funneling, bay effects, etc.). HURCON definitions are as follows:

7.8.1. HURCON 5: 50 kts/ 58 miles per hour (mph) winds or higher within 96 hours.

7.8.2. HURCON 4: 50 kts/58 miles per hour (mph) winds or higher within 72 hours.

7.8.3. HURCON 3: 50 kts/58 miles per hour (mph) winds or higher within 48 hours.

7.8.4. HURCON 2: 50 kts/58 miles per hour (mph) winds or higher within 24 hours.

7.8.5. HURCON 1: 50 kts/58 miles per hour (mph) winds or higher within 12 hours.

7.8.5.1. HURCON 1C: 34 kts/40 miles per hour (mph) or higher are occurring. All outdoor activities are discontinued, except for urgent military missions.

7.8.5.2. HURCON 1E: All outside activity is strictly prohibited.

7.8.5.3. HURCON 1R: Life-threatening storm hazards have passed but damage may persist, and only emergency responders and damage assessment personnel are released to move about.

7.8.6. BOS/WSO /26 OWS will augment the NHC forecasts to include the following:

7.8.6.1. Onset of crosswinds and weather phenomena or conditions expected that could prevent evacuation of aircraft.

7.8.6.2. BOS/WSO provides forecasts to the DRF/commanders and recommends HURCON levels to the installation commander, who is the declaring authority. See 81 TRW PLAN 10-2, IEMP for more hurricane information.

7.8.6.3. BOS/WSO will notify the installation commander when a tropical system is developing in the Gulf of Mexico or forecast to move into the Gulf. Based on the official forecast and the situation, BOS/WSO can recommend the DRF to convene.

7.8.7. Disseminating Hurricane Information. When an active tropical system is within or forecast to be within 960NM of KEESLER AFB and has the potential for significant impact on KEESLER AFB, or by direction of 81 TRW and BOS/WSO leadership. BOS/WSO will issue the Hurricane Bulletin via email and posted to the 81OSF-Weather SharePoint site. At a minimum, the Hurricane Bulletin will be sent to the Keesler Hurricane Bulletin e-mail address list. Individual agencies will ensure the bulletin is disseminated throughout their organizations.

7.9. Chemical, Biological, Radiological Nuclear, and High Yield Explosive Data (CBRNE). The 81 CES/CEF (Fire Protection Flight) is the primary unit for providing CBRNE support. The CDM and EDM are used to determine the spread CBRNE of agents that are released. The EDM bulletin is produced by 557th WW and is used for forecasting threat areas for fallout from nuclear detonation. The CDN is available at both 557th WW and the 26 OWS. Per DAFI 10-2501 BOS/WSO is responsible for providing only meteorological data towards the calculation of Toxic Corridor products. The 26 OWS may also supplement BOS/WSO with this weather data-only support, if required.

7.10. Severe Weather Damage Reporting/OPREPs. IAW DAFMAN 15-129 and AFMAN 10-206, *Operational Reporting (OPREP)*, BOS/WSO will query base cooperative weather watch agencies (SF, Fire Dept, etc.), local spotters and online damage assessment data sources (NWS, local news, local law enforcement, etc.) to collect the area impacts of severe weather events (on base or in the immediate surrounding areas) as time permits after a severe weather occurrence. BOS/WSO will coordinate with CP for weather related Operational Report 3 (OPREP-3) and provide the CP with any pertinent base information. The CP, in turn, will provide BOS/WSO with an unclassified copy of any weather related OPREPs. BOS/WSO will provide damage reports and OPREP-3s to the 26 OWS and AETC/A30W and CP will provide OPREP-3s to Higher Headquarters within 15 mins of the event/incident. **Note:** The 81 TRW/CC is the releasing authority for OPREP-3s.

7.10.1. Information required to be provided by BOS/WSO to the CP includes:

7.10.1.1. Date/time and type of actual severe weather experienced.

7.10.1.2. Forecast valid at the time of occurrence.

7.10.1.3. Any watches, warnings issued (include actual and desired lead times).

7.10.1.4. Operational status of meteorological equipment (radar, wind sensors, etc.) at both the WSO and the 26 OWS.

7.10.1.5. Other pertinent information (damage reports, visual/radar tornadic confirmation, etc.).

Chapter 8

WEATHER INFORMATION DISSEMINATION

8.1. General. Forecasts, observations, and weather alerts (warnings, watches, and advisories) are useful only if the customers have access to them. Timely and efficient dissemination of weather information is crucial to the success of the wing mission and resource protection. Most organizations will receive this information via our primary dissemination system. Some organizations will receive courtesy calls, however, DAFMAN 15-129 restricts the number of organizations BOS/WSO can directly notify (via telephone) in order to focus on incoming weather phenomena. This chapter describes the weather dissemination systems, dissemination procedures and backup systems and procedures.

8.2. Dissemination Systems and Backups.

8.2.1. Automated Dissemination System. In this publication, Automated Dissemination System (ADS) refers to any Air Force or Army accredited dissemination system (e.g., Joint Environmental Toolkit (JET)). The ADS is the primary method for dissemination of observations, forecasts, warnings, watches, and advisories. Agencies without access to the ADS will receive critical weather information through their unit's own dissemination system (e.g., Pyramid Alert Recall, Unit Control Center, etc.). In the event that the ADS is inoperable BOS/WSO will use the backup procedures listed in section 8.3.

8.2.2. Local Area Network (LAN). The LAN is the primary method used for gathering weather information. Additionally, it is used as the backup system for disseminating (long- line only) weather observations. In the event of a LAN outage, BOS/WSO has obtained the use of a cable modem through the local cable company to link to the internet. When using the cable modem, BOS/WSO will be limited on its use of some military internet sites and will not have email services available.

8.2.3. Hotlines. Hotlines have been installed between BOS/WSO and other agencies for the sole purpose of rapidly exchanging weather data. Normal phone lines will be used in the event of a hotline outage. **Table 8.1**; outlines hotlines that are installed between BOS/WSO and the following agencies (DSN Prefix 597, Comm. 228-377):

Table 8.1. Hotlines BOS/WSO.

Control Tower	228-377-3227
Command Post	228-377-4330
81TRW Commander	228-377-8100

8.2.4. Climatic Data Requests. Upon Request from an authorized agency (as deemed appropriate by station management), weather personnel will provide climatology data for KEESLER AFB and other locations. Request may be made by phone, e-mail, letter, or in person. Monthly climatology will be made available to the following agencies in **Table 8.2**.

Table 8.2. Monthly Climatology Distribution/Authorization.

14th Weather Squadron	Harrison County Wastewater and Solid Waste Management
81 FSS/FSCO	Mississippi Department of Marine Resources
53 WRS, 815 AS	City of Biloxi Engineering
BOS/SVC	81 TRW/SE
81 MSDS/SGSM	81 MSG/PMO
A&G CRTC – Gulfport/CE	81 CES/FD
81 CONS/LGCA	2AF/JA
Navy Construction Office Gulfport	81 TRW/JA
BOS/CE and CEV	

8.3. Dissemination and Backup Procedures.

8.3.1. TAFs. TAFs will be entered and disseminated through the ADS. If the ADS is out of service, then BOS/WSO will disseminate the information locally via standard telephone lines to the CP and ATC Tower, and Airfield Management.

8.3.2. Observations Procedures. Observations will be taken and disseminated through the ADS. When the ADS is out of service, observations will be disseminated long-line through AFW-WEBS using the LAN. If both the ADS and the LAN are out of service, long-line dissemination will also be through AFW-WEBS using a dedicated cable connection not associated with the LAN. If all long-line transmission resources are out of service, observations will be called out via telephone (voice relay) to any other weather station or supporting hub (Barksdale AFB, LA) for long-line transmission. Locally, observations will be relayed by BOS/WSO to the following organizations (noted on a local dissemination log) in order of priority when the ADS is out of service:

8.3.2.1. Tower (Hotline or 228-377-8877).

8.3.2.2. Keesler CP (Hotline or 228-377-4330)

8.3.2.3. Gulfport Tower (228-864-3760).

8.3.2.4. WBGTI. The WBGTI will be disseminated locally by the Bio-environmental Flight on duty days or the Blake Fitness Center on non-duty days.

8.3.3. **Weather Watches, Warnings, and Advisories.** Weather watches, warnings, and advisories will be disseminated via the ADS. When warnings are issued, the 26 OWS or BOS/WSO, they will give a courtesy call to the CP. BOS/WSO will also give the KEESLER AFB Tower a courtesy call to ensure receipt. If the ADS is out of service, the 26 OWS and/or BOS/WSO will contact the same organizations via telephone. Further dissemination by BOS/WSO, Airfield Management, and the Command Post is as follows:

8.3.3.1. 81 Training Group Commander (81 TRG/CC) Notification. BOS/WSO will notify by phone the 81 TRG/CC of all lighting watches and warnings, Monday thru Friday (excluding NTDs) between the hours of 0600L and 1800L. BOS/WSO will also notify the 81 TRG/Military Training Office (MTO). Military Training Leaders (MTLs) will be notified of all lightning watches and warnings (excluding NTDs) between 0500L-0700L.

8.3.3.2. 81 OSF/CC Commander Notification**.** Weather Operations will ensure the 81 OSF/CC is notified after airfield hours when Severe Thunderstorm watches and warnings are issued as they may require the recall of Transient Alert contractors.

8.3.3.3. Watch, Warning, and Advisory Notifications. Watches, warnings, and advisories will be further disseminated by the CP and Airfield Management according to **table 8.3** and **table 8.4**. Units listed will develop methods to notify subordinate agencies. Groups should use their 24-hour operations to recall other key personnel/units that are needed to implement resource protection plans.

8.3.3.4. Airfield Management Operations. The Keesler CP will relay all weather warnings via AtHoc. When required as a backup to AtHoc, Airfield Management Operations will relay via the Secondary Crash Net, or request the Keesler CP to activate the KAN. The applicable organizations are outlined in **Table 8.3** (if necessary, landlines are used for backup purposes):

Table 8.3. Notifications via the Secondary Crash Net.

Organizations Notified by Airfield Management Operations	
Organization	**Contact Information**
Fire Department	Hotline or 228-377-3333
Emergency Room	Hotline or 228-377-0500
81st Security Forces Squadron	Hotline or 228-377-3040
53d Weather Reconnaissance Squadron	228-377-2409
403 WG Maintenance Operations Squadron	Hotline or 228-377-4220
Command Post	Hotline or 228-377-4330
Safety (Normal Duty Hours)	228-377-2910
CE Readiness (Normal Duty Hours)	228-377-2655
81st Mission Support Group Commander (Normal Duty Hours)	228-377-2510
Public Affairs (Normal Duty Hours)	228-377-2783
Flight Medicine (Normal Duty Hours)	228-377-6668
Weather	228-377-3305
Alternate Security Forces Squadron	228-377-3065
Transient Alert	377-2521

8.3.3.5. Command Post. The Keesler CP disseminates all weather watches and warnings via AtHoc. AtHoc is the primary system for disseminating weather notifications. The KAN will be activated as a backup to Secondary Crash and/or as requested by AMOPS. If AtHoc is not used, calls must be placed to the agencies with an asterisk (*). In the case of Airfield Management, notification is provided using both the KAN and by the ADS. **Table 8.4** outlines the organizations that are on the KAN: Hotline or 377-2521.

Table 8.4. Notifications via the Keesler Alert Net.

Upon receipt of a Weather Watch, Warning or Advisory, the Command Post will notify, via the KAN, the organizations listed below.		
Organization	**Phone Number**	
Security Forces	228-377-3040	*
Fire Department	228-377-3333	*
Transient Alert	228-377-2521	*
Medical Group (ER)	228-376-0500 / 4036	*
CES (EMCS)	228-377-4179 / 4936	*
403 WG MOC*	228-377-4220	*
Airfield Management Operations	228-377-2120 / 2123	*
Fuels Management	228-377-2964	*
81 MSG	228-377-2510	*
Safety	228-377-2910 / 7023	*
Public Affairs	228-377-2783 / 3901	*

8.3.3.6. Severe Weather. When any type of severe weather watch or warning is issued, CP notifies the base leadership (wing and group commanders) via AtHoc. This system notifies base personnel via pop-ups on their computers, emails, text messages, phone calls and mobile application (as applicable).

8.3.3.7. Tornado Warnings. The Keesler CP has the primary responsibility for sounding the base siren in the event of a Tornado Warning issuance. This will be immediately followed up with activation of AtHoc to provide additional warning to the base populace.

Chapter 9

WEATHER EQUIPMENT

9.1. General. This chapter provides a brief description of the meteorological and communications equipment used by BOS/WSO. Additionally, it provides information on backup systems, maintenance and restoring priorities.

9.2. Weather Equipment. BOS/WSO uses a wide range of equipment to determine the current state of the atmosphere. These critical systems are used continuously to provide customers the most timely, accurate and relevant weather intelligence possible. KEESLER AFB has a single-instrument runway with exception of the dual sensor wind and cloud measuring equipment. Readouts for the observing equipment are located in the BOS/WSO area.

9.2.1. AN/FMQ-19 Automated Meteorological Observing System (AMOS). The AN/FMQ-19 AMOS is the primary airfield weather observation system capable of measuring wind speed/direction, temperature/dew point, pressure, prevailing visibility/present weather, cloud ceiling heights, precipitation amounts, freezing rain detection and lightning strike detection. Sensors are located on the airfield and information is displayed in BOS/WSO. Backup for the AN/FMQ-19 AMOS is the Kestrel (wind speed, temperature/dew point, pressure), AFW-WEBS (lightning strike detection) and a Weather Forecaster (clouds, visibility, atmospheric phenomena).

9.2.2. Gibson/Ridge Radar Program. The Gibson/Ridge Radar Program is a hardware/software system designed for the display and interrogation of Doppler radar imagery on a Graphical User Interface (GUI). The Gibson/Ridge Radar Program allows Weather Forecasters to display, locate and interrogate severe thunderstorms as well as tornadic activity. The Internet provides BOS/WSO backup radar information. Control of the Gibson/Ridge Radar Program sites belongs to the National Weather Service. Backup for the Gibson/Ridge Radar Program is other reliable internet sites as applicable. The 26 OWS will also provide backup support when needed.

9.2.3. Automated Dissemination System (ADS)**.** As discussed in **Chapter 8**, the ADS is the primary system for dissemination and is located in bldg 0233, room 109.

9.2.4. PMSV Radio**.** The PMSV Radio (267.4 MHz) allows BOS/WSO to communicate with aircrews, both on the ground and in the air, as well as tower personnel. If the PMSV is out, Contact the ATC Tower and let them know that our PMSV is out. Ask them to inform aircrews trying to contact Keesler Metro that PMSV is out, and they will need to contact another location for assistance. Aircrews can contact the 26 OWS via phone patch (DSN 331-2651 or COMM 318-529-2651) to get weather data.

9.2.5. Phones/Hotlines. Phones and hotlines primarily serve as a backup system as well as for passing along Critical and time sensitive information rapidly.

9.2.6. LAN. BOS/WSO relies heavily on the local area network to improve the timeliness and accuracy of weather intelligence to our customers. As a backup to the LAN, BOS/WSO has obtained a Cable Modem from a local cable company.

9.2.7. Kestrel. The Kestrel is a hand held measuring device that measures Barometric Pressure, Altitude, Density Altitude, Temperature, Humidity, Wind Speed, Wind Chill, Dew Point, Wet Bulb, and Heat Index. This instrument is used as the primary back up to the AN/FMQ-19 AMOS in the event of an outage.

9.2.8. Lightning detection and prediction is accomplished by Forecasters by use of officially approved products and techniques. Among these are the AN/FMQ-19, official Air Force websites, and Doppler Weather Radar.

9.2.9. Automated Dissemination System (ADS). An ADS has been installed and provides the users with a single, fully integrated system of weather analyses, forecasting, and dissemination capabilities.

9.3. Maintenance. All equipment requires some sort of maintenance from time-to-time. Weather data is obtained via leased commercial telephone lines, the Internet and satellite download for ingest into the various weather systems.

9.3.1. The 81 OWS/RAWS will maintain the following weather communications equipment around the clock. Response to weather communication equipment outages during other than normal duty hours will be based on mission impact. Communication systems that interface with commercial telephone companies will be coordinated and tracked by the 81 OSF/RAWS.

9.3.1.1. PMSV.

9.3.1.2. Class A/local telephone lines and phone lines connecting to weather equipment on the airfield.

9.3.1.3. ADS communication lines.

9.3.1.4. AN/FMQ-19 AMOS.

9.3.2. 81OSF/RAWS will respond to weather equipment or communication outages that prevent mission accomplishment within 30 minutes during normal duty hours and 45 minutes after duty hours, weekends, and holidays. Equipment restoral priorities, reasoning and response times are listed in **Table 9.1,** in order of priority, highest to lowest. Deviations are authorized based on existing and forecast weather conditions as specified by a Weather Forecaster or BOS/WSO management.

Table 9.1. Equipment Restoral Priorities.

EQUIPMENT	REASONING	RESPONSE TIME
AN/FMQ-19 AMOS	Observes flight line weather conditions. Critical to flight safety.	30 Minutes (duty hours) 45 minutes (after duty hours)

Chapter 10

26 OWS/BOS/WSO BACKUP SUPPORT

10.1. General. There are dozens of scenarios that could cause an interruption of service from either the 26 OWS or Weather Operations. This chapter briefly describes how weather services will be provided should any such events occur.

10.2. 26OWS.

10.2.1. 26OWS Backup Weather Procedures for BOS/WSO. Per Keesler AFB/26 OWS Datasheet when weather operations are interrupted (e.g., power outage, natural disaster, etc.) resource protection, weather watch and warning responsibility and transient flight briefing will be transferred to the 26 OWS until BOS/WSO is prepared to resume operations.

10.2.2. 26OWS Long-term Backup Weather Procedures. Per Datasheet, when the interruption in weather operations at the 26 OWS is expected to exceed 72 hours, weather responsibility for KEESLER AFB will transfer IAW DAFMAN 15-129. The 26 OWS will relocate to either 557th WW or to another Continental United States (CONUS) OWS. Once in place and operational, the 26 OWS will resume support for KEESLER AFB to the greatest extent possible as resources allow.

10.3. Weather Operations. For standard station evacuations, support will resume from our alternate weather operations site with the 26 OWS assuming responsibilities during the interim period. For longer interruptions, BOS/WSO will coordinate required support with other organizations. Per Datasheet, the 26 OWS will not normally produce and disseminate tactical-level weather products for KEESLER AFB mission execution. However, when operational constraints prevent WSO from accomplishing the task, the 26 OWS may produce and disseminate selected tactical-level weather products (Web-Based FAMEFs, verbal discussions, emailed products, etc.) on a short-term basis. BOS/WSO will coordinate the transfer of this responsibility in advance with the 26 OWS.

Chapter 11

RECIPROCAL SUPPORT

11.1. General. The WSO requires daily support in order to best accomplish its mission. This chapter outlines the reciprocal support required between WSO and other units on Keesler AFB. Per DAFMAN 15-129, support to BOS/WSO by other local agencies that are mandated by USAF or other local directives are not included in this chapter.

11.2. Keesler Command Post (Keesler CP). Disseminate weather advisories, watches, and warnings according to 81 TRW checklists. The Keesler CP will:

11.2.1. Obtain information from BOS/WSO management for input into OPREP-3 and BEELINE reports, which involve severe weather, aircraft mishaps or natural disasters at Keesler AFB.

11.2.2. When BOS/WSO is closed, contact the Keesler standby Weather Forecaster, and when the 26 OWS calls with a severe watch or warning, the standby forecaster will come in. BOS/WSO will provide Keesler CP a standby roster each time they close.

11.2.3. Provide classified storage for BOS/WSO classified materials.

11.3. 81OSF Airfield Management (81 OSF/OSAM). During airfield operating hours, will:

11.3.1. Notify BOS/WSO of all aircraft ground emergencies, in-flight emergencies, and aircraft accidents via the secondary crash phone.

11.3.2. Provide flight information to BOS/WSO on DVs, transient aircraft schedules and changes to local aircrew flight plans/schedules.

11.3.3. 81 OSF/CC will forward quarterly Airfield Operations Board meeting minutes/agendas to WSO. Weather and Performance Management Office representatives are expected to attend the Airfield Operations Board meetings as much as their other duties permit.

11.3.4. Supply Keesler BOS/WSO with required Flight Information Publication (FLIP) and FLIP-related products. Updates weather information in the FLIP as initiated by Keesler Weather Operations.

11.3.5. Include weather support information in NOTAMs issued when airfield sets other than normal duty hours, etc.

11.3.6. Provide Runway Surface Condition to the Weather Forecaster.

11.4. 81OSF Control Tower (81 OSF/OSAT) will:

11.4.1. Conduct daily operational checks of the PMSV radio (equipment permitting) on days the airfield is open, preferably between 0745-0800L; on weekends between 1045-1100L.

11.4.2. Notify BOS/WSO of active runway changes.

11.4.3. Participate in a cooperative weather watch as stated in **paragraph 4.7**.

11.4.4. Notify BOS/WSO whenever the tower's ADS terminal or AN/FMQ-19 AMOS terminal become inoperative or returns to service.

11.4.5. IAW KEESLER AFBI 13-204, Airfield Operations and Local Flying Procedures, make a broadcast over the tower, ground, and ramp net frequencies for a lightning warning within five (5) NM of Keesler. The tower will also broadcast the lightning warning on the Automatic Terminal Information Service (ATIS).

11.5. 81OWF/RAWS. Provides and arranges for maintenance of weather and communications equipment and provides a single 24-hour point of contact for logging out equipment listed in **paragraph 9.3**. RAWS will periodically update Weather Forecasters on the status of open maintenance actions. The 81 OSF/RAWS will also:

11.5.1. Restore equipment in the order listed in **paragraph 9.3** and respond to weather equipment outages that prevent mission accomplishment within 30 minutes during normal duty hours and 45 minutes after duty hours or on weekends and holidays. Direct contract support when required, i.e., interface with commercial telephone lines.

11.5.2. Perform routine maintenance requests in a timely manner as the maintenance workload permits. Maintain a periodic Maintenance Inspection log and schedule on weather equipment.

11.5.3. Minimize network traffic upon 81 TRW/CC request in order to provide mission critical network access to Weather Forecaster operational computer systems.

11.5.4. Provide assistance with planning weather system upgrades that interface with the base LAN.

11.5.5. Provide assistance/maintenance of pager notification system for units using it to disseminate weather warnings. Notify units when the system is not working properly.

11.5.6. Host the Keesler web site where the weather homepage resides and provide a means for public notification and viewing of weather products.

11.6. 81TRW Safety (81 TRW/SE). Notify BOS/WSO of all mishaps in which weather or weather services could have been a factor.

11.6.1. Include weather support procedures in the annual flying safety program.

11.6.2. Inspect 81 TRW organizations and tenant units for compliance with any weather-related Air Force Occupational Safety and Health standards.

11.7. 403Wing (403WG) (53WRS/815AS). Inform BOS/WSO of declared alternate airfields and provide BOS/WSO with PIREPs received from unit aircraft.

11.7.1. Pass updated weather information to respective aircrews IAW AFI 11-418.

11.7.2. Provide BOS/WSO with feedback IAW **paragraph 1.9** of this instruction.

11.8. BOS Civil Engineering (BOS/CE). Ensure BOS/WSO has an auto start generator to provide emergency power to Bldg 0233 in the event of a power outage. Provide monthly generator maintenance checks and train Weather Forecasters on the use of the emergency generator.

11.8.1. Notify BOS/WSO of property damage due to weather phenomena.

11.8.2. When notified of air conditioning problems in Bldg 0233, room 109, BOS/CE will respond on an “Emergency” basis. When existing equipment cannot be repaired in a timely manner, BOS/CE will attempt to provide an alternative method of cooling.

11.8.3. Ensure any contractor support requiring weather notifications receive said notification in a timely manner.

11.9. 81Security Forces Squadron (81 SFS). Urgently notifies BOS/WSO when observing adverse weather phenomena or conditions as part of the Cooperative Weather Watch outlined in **paragraph 4.7** of this instruction.

11.10. 81TRW Wing Inspection Team. Assist BOS/WSO with conducting exercises.

11.11. Flying Units Located on Keesler AFB. Will provide BOS/WSO at least 72-hours' notice when special weather support is required for off station or mass briefings, etc.

11.11.1. Will coordinate two weeks in advance for exercise related weather support and special briefing such as IRC, Quarterly SOF/DUTY OFFICER Briefings, Safety Briefings, etc.

11.11.2. Provide PIREPS IAW with criteria listed in **paragraph 4.7.4.3**; of this instruction.

11.11.3. Provide BOS/WSO with accurate weekly/daily flying schedules upon publication and inform BOS/WSO in a timely manner of mission cancellations to eliminate preparing unneeded weather briefings.

11.11.3.1. Provide BOS/WSO and/or 26 OWS Weather Forecasters two hours lead time for preparation of flight weather briefings.

11.11.3.2. For briefings required outside the normal duty hours of BOS/WSO (to include preparation time) contact the 26 OWS (DSN ***331-2651***) to set up a briefing.

11.12. 81st Training Group (81 TRG).

11.12.1. The 81st Training Group, Military Training Office (81 TRG/MTO) will provide parade and drill down information and schedule changes to BOS/WSO to meet responsibilities set forth in DAFPAM 34-1203, KEESLER AFB Supplement, *Drill and Ceremonies*.

11.12.2. Notify BOS/WSO of special events that require added METWATCH for weather criteria that may preclude the event. During BOS/WSO non-duty hours, contact the 26 OWS at DSN 331-2651 or 331-2622 for required weather support.

11.12.3. The 81 TRG/MTO establishes procedures for NPS Airmen when troop movements halt during hazardous weather conditions and coordinates with host base transportation to provide transportation.

11.13. 335th Training Squadron/Weather Training Flight (335 TRS/UOA). 335 TRS/UOA will allow the WSO to use its facility as the AOL in the event that Airfield Management needs to be evacuated. They will assign a cubical to WSO to use and allow the staging of equipment and supplies for storage and easy access.

BILLY E. POPE JR, Colonel, USAF
Commander, 81st Training Wing

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

Keesler AFB Installation *Emergency Management Plan 10-2,* 22 May 2022

AETCI 36-2216, *Administration of Military Standards and Discipline Training,* 17 April 2014

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DAFI 10-2501, *Emergency Management Program,* 15 Oct 2023 AFMAN 11-2C-130Jv3, *C-130 Operations Procedures,* 31 July 2023

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AFMAN 11-202v3, *AETC Supplement General Flight Rules,* 21 September 2022

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AFMAN 13-204v3, *Air Traffic Control,* 21 July 2020

AFI 33-322, *Records Management and Information Governance Program,* 27 July 2021

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DAFI 10-2503 *Operations in a Chemical, Biological, Radiological, Nuclear CBRN Program,* 5 October 2023

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AFMAN 15-111, *Surface Weather Observations,* 11 March 2019

AFMAN 15-124*, Meteorological Codes,* 15 January 2019

DAFMAN 15-129, *Air and Space Weather Operations* 6 September 2023

DAFPD 15-1*, Air Force Weather Operations,* 13 November 2019

26OWS *and BOS/WSO Datasheet*

FAA Order 7110.65AA, *Air Traffic Control,* 5 October 2023

Keesler AFB, (IDP) *Installation Deployment Plan,* 11 October 2016\

Prescribed Forms

None

Adopted Forms

AF Form 3803, *Federal Meteorological Surface Weather Observations (METAR/SPECI)*

DAF Form 3806, *Weather Watch Advisory Log*

DAF Form 3807, *Watch/Warning Notification and Verification* AF Form 847, *Recommendation for Change of Publication*

DD Form 175-1, *Flight Weather Briefing*.

Abbreviations and Acronyms

AAI—Aircraft Accident Investigation

ADS—Automated Dissemination System

AETC—Air Education and Training Command

AETCI—Air Education and Training Command Instruction

AF—Air Force

AFB—Air Force Base

AFI—Air Force Instruction

AFMAN—Air Force Manual

AFPD—Air Force Policy Directive

AFT—After

AFW—Air Force Weather

AFW-WEBS—Air Force Weather Web Based System

AFWWS—Air Force Weather Weapons System

AMD—Amendment

AMOS—Automated Meteorological Observing System

AGE—Aerospace Ground Equipment

AGL—Above Ground Level

ALSTG—Altimeter Setting

AOL—Alternate Operating Location

AOR—Area of Responsibility

APOD—Aerial Port of Debarkation

APOE—Aerial Port of Embarkation

ATC—Air Traffic Control

AWN—Automated Weather Network

BAFB—Barksdale Air Force Base

BWW—Basic Weather Watch

CDM—Chemical Downwind Message

CI—Chill Index

CIG—Ceiling

CONUS—Continental United States

CP—Command Post

CS—Communications Squadron

CSAF—Chief of Staff of the Air Force

CT—Central Time

CWW—Continuous Weather Watch

DCC—Deployment Control Center

Ddhh—Day of Month/Hour in Greenwich Mean Time

DEST—Final Destination

DLT—Desired Lead-Time

DP—Disaster Preparedness

DRF—Disaster Response Force

EDM—Effective Downwind Message

EM—Emergency Management

EWO—Emergency War Order

FAMEF—Flying Area Mission Execution Forecast

FC—Funnel Cloud

FLIP—Flight Information Publication

FSTR—Full Spectrum Threat Response

FWA—Forecast Weather Advisory

FWB—Flight Weather Briefing

GPS—Global Positioning System

GR—Hail of 1/4 inch or more

HF—High Frequency

IAW—In Accordance With

IDO—Installation Deployment Officer

IEMP—Installation Emergency Management Plan

IRC—Instrument Refresher Course

IRP—Instrument Refresher Program

ITS—Index of Thermal Stress

JET—Joint Environmental Toolkit

JFCOM—Joint Forces Command

KEESLER AFB—Keesler Air Force Base

KEESLER AFBI—Keesler Air Force Base Instruction

KAN—Keesler Alert Network

KBIX—Keesler AFB, MS

KGPT—Gulfport, MS

KJAN—Jackson, MS

KBHM—Mobile, AL

KMXF—Maxwell AFB, AL

KT—Knots

LAN—Local Area Network

LDS—Lightning Detection System

LLWS—Low Level Wind Shear

MEDEVAC—Medical Evacuation

MEF—Mission Execution Forecast

MEFP—Mission Execution Forecast Process

METAR—Routine Meteorological Observation Report

METNAV—Meteorological and Navigational

METSAT—Meteorological Satellite

METWATCH—Meteorological Watch

MOA—Memorandum of Agreement

MOS—Manual Observing System

MRO—Medical Response Option

NCOIC—Non-Commissioned Officer in Charge

NEXRAD—Next Generation Radar

NHC—National Hurricane Center

NM—Nautical Mile

NOAA—National Oceanic and Atmospheric Administration

NOTAMS—Notice to Airmen

NTD—Non-Training Day

NWS—National Weather Service

OIC—Officer in Charge

OID—Operator Interface Device

OPREP—Operational Report

OSF—Operations Support Flight

OWA—Observed Weather Advisory

OWL—Operational Weather Limiters

OWS—Operational Weather Squadron

PIREP—Pilot Report

PMSV—Pilot to Metro Service

POC—Point of Contact

QA—Quality Assurance

RVR—Runway Visual Range

SATCOM—Satellite Communications

SF—Security Forces

SFMR—Stepped Frequency Microwave Radiometer

SM—Statute Mile

SOF—Supervisor of Flying

SOP—Standard Operating Procedures

SPECI—Special

SWAP—Severe Weather Action Procedures

SWAT—Severe Weather Action Team

TAF—Terminal Aerodrome Forecast

TBD—To Be Determined

TPC—Tropical Prediction Center

TRW—Training Wing

UA—Text Element Indicator used in routine PIREPs

UHF—Ultra High Frequency

UTA—Unit Training Assembly

UUA—Text Element Indicator used in urgent PIREPs

VA—Volcanic Ashe

VHF—Very High Frequency

VIS—Visibility

VSAT—Very Small Aperture Terminal

WBGTI—Wet Bulb Global Temperature Index

Office Symbols

53 WRS—Weather Reconnaissance Squadron

81 CES/CEF—Fire Protection Flight

81 CONS—Contracting Squadron

81 FSS—Force Support Squadron

81 OSF—Operations Support Flight

81 OSF/AFM—OSF, Airfield Management

81 OSF/OSW—OSF, Weather

81 TRW—81st Training Wing

81 TRW/CC—Commander

81 TRW/CD—Deputy Commander

81 TRW/PA—Public Affairs

815 AS—Airlift Squadron

A & G CRTC—Combat Readiness Training Center

BOS/WSO—Weather Services Office

Terms

Actual Lead-time—The elapsed time between issue time of a watch, warning, or advisory and the first occurrence of the event.

Advisory—A formal notice of a weather condition that will or is impacting flight operations.

Aircraft Mishap—Term used to denote any event resulting in damage to, or destruction of any aircraft to include lightning strikes, inadvertent departure from the paved runway or taxiway surface, aircraft or Aerospace Ground Equipment (AGE) fires, and forced landings due to in-flight emergencies.

Alternate Operating Location (AOL)—The location to which BOS/WSO will move in the event that Airfield Management is evacuated.

Amendment (AMD)—Used as a message modifier when transmitting an aerodrome forecast amendment.

Automated Dissemination System (ADS)—Refers to any Air Force or Army accredited dissemination system (e.g., Joint Environmental Toolkit (JET)).

AN/FMQ—19 AMOS – Automated Meteorological Observing System used to measure atmospheric conditions. System can measure: wind speeds and direction, temperature, dew point, visibility, pressure, runway visual range and some weather phenomenon.

Basic Weather Watch (BWW)—A method of observing, recording, and disseminating significant changes in weather conditions to essential customers.

Climatology—The historical records of weather conditions measured or observed at a specific location is known as climatology. Some data go back over 100 years, but generally; a 10- to 25-year history is more common. Climatology is useful in planning operations beyond five (5) to seven (7) days. It usually describes the average (or mean) conditions such as high and low temperatures and extremes.

Continuous Weather Watch (CWW)—Weather Forecasters will monitor weather conditions continuously and perform no other significant duties. In addition to taking METARs, Weather Forecasters will take and disseminate observations as conditions occur that meet SPECI observation criteria.

Desired Lead-time—The amount of advance notice a supported agency desires before the onset of a particular weather phenomenon.

Eyes Forward—Weather Forecasters are the eyes forward for the forecasters in the Operational Weather Squadron (OWS) and integrate weather radar data, meteorological satellite imagery, lightning detection readouts, and non-standard weather data systems (vertical profilers, mesonet data, etc.) to create an integrated weather picture and near-term trend forecasts for the OWS. Eyes forward yields meaningful meteorological information not contained in coded observation to the servicing OWS and is an integral part of the meteorological watch for an installation or contingency operating location.

Weather Flights—Provide direct or continuing support to a mission set and determine both environmental threats and effective decision points to inject weather into the planning and execution process of the mission. This allows for development of courses of action during the planning process to mitigate these threats

Flying Area Mission Execution Forecast (FAMEF)—A customized weather product providing terrestrial and space weather data and forecasts for a specific mission, or set of missions. It fully integrates aerospace weather with the customer's tactics, weapon systems, environmental sensitivities of equipment, and other operational requirements.

Forecast Weather Advisory—A weather advisory issued when the customer requires advance notification of an impending weather condition with sufficient time to allow for protective actions.

Gibson/Ridge Radar Program—A suite of data processing and display programs developed to view weather radar data.

Issue Time—The time when an agency is notified of a watch, warning, or advisory. When more than one agency is notified, the issue time is the time the last agency is notified. Follow-up notifications are not considered when determining issue time.

MEF—Mission tailored environmental information describing a specific impact to an operational mission. BOS/WSO will conduct deliberate forecast processes to develop, deliver, monitor, and amend mission execution forecasts by fusing perishable data with operational and strategic level weather forecast products into decision quality information for an operational end user.

METAR Observation—Meteorological Aviation Report: A routine scheduled surface weather observation. It contains a report of wind, visibility, runway visual range, present weather, sky condition, temperature, dew point, and altimeter setting. In addition, significant remarks are appended to the METAR observation.

METWATCH—Monitoring aerospace weather for a route, area, or terminal and advising concerned organizations when phenomena that could affect their operations or pose a hazard to life or property are occurring or about to occur.

MISSIONWATCH—The monitoring of aerospace weather for a specific mission (i.e., ground, air, or space) and informing supported agencies when unforecasted mission-limiting phenomena could impact operations.

Observed Weather Advisory—A weather advisory issued when a particular weather event first occurs and the customer does not require advanced notification of the observed weather phenomena.

Operational Weather Squadron—An organization comprised of management, forecasters, and training personnel responsible for providing regional weather support. Their mission is to produce theater-scale tailored weather forecast products and services to customers within their area of responsibility.

Pilot Report (PIREP)—A report of in-flight weather provided by an aircrew member.

Severe Thunderstorm—A thunderstorm that produces hail greater than or equal to ¾ inch diameter and/or surface wind greater than or equal to 50 knots.

Severe Weather—Any weather condition that poses a hazard to property or life.

SPECI Observation—An unscheduled observation taken when significant changes in weather elements meet special criteria. All SPECIs shall be made as soon as possible after the relevant criteria are observed.

Terminal Aerodrome Forecast (TAF) —TAF is composed of required weather elements for Keesler AFB airfield and covers a 30-hour period. Forecast elements in the body of the forecast text refer to the area within 5NM of the center of the aerodrome complex. Operationally significant elements outside this area are included in remarks (e.g., TS OMTNS or VCTS). The term VC (vicinity) refers to the area between 5NM and 10NM of the aerodrome complex.

Tornado Watch—Conditions are conducive to the development of tornadoes in and close to the watch area.

Tornado Warning—A tornado has actually been sighted by spotters or indicated on radar and is occurring or imminent in the warning area.

Weather Advisory—A special notice provided to a supported agency when an established weather condition that could affect its operation is occurring or is expected to occur.

Weather Warning—A special notice provided to a supported agency when an established weather condition of such intensity as to affect operations, pose a hazard to life or property, and requires protective action, is occurring or is expected to occur.

Weather Watch—A special notice provided to supported customers that alerts them of a potential for weather conditions of such intensity as to pose a hazard to life or property for which the customer must take protective action.

Attachment 2

LOCAL FLYING AREA MAPS

Figure A2.1. 815th Airlift Squadron.

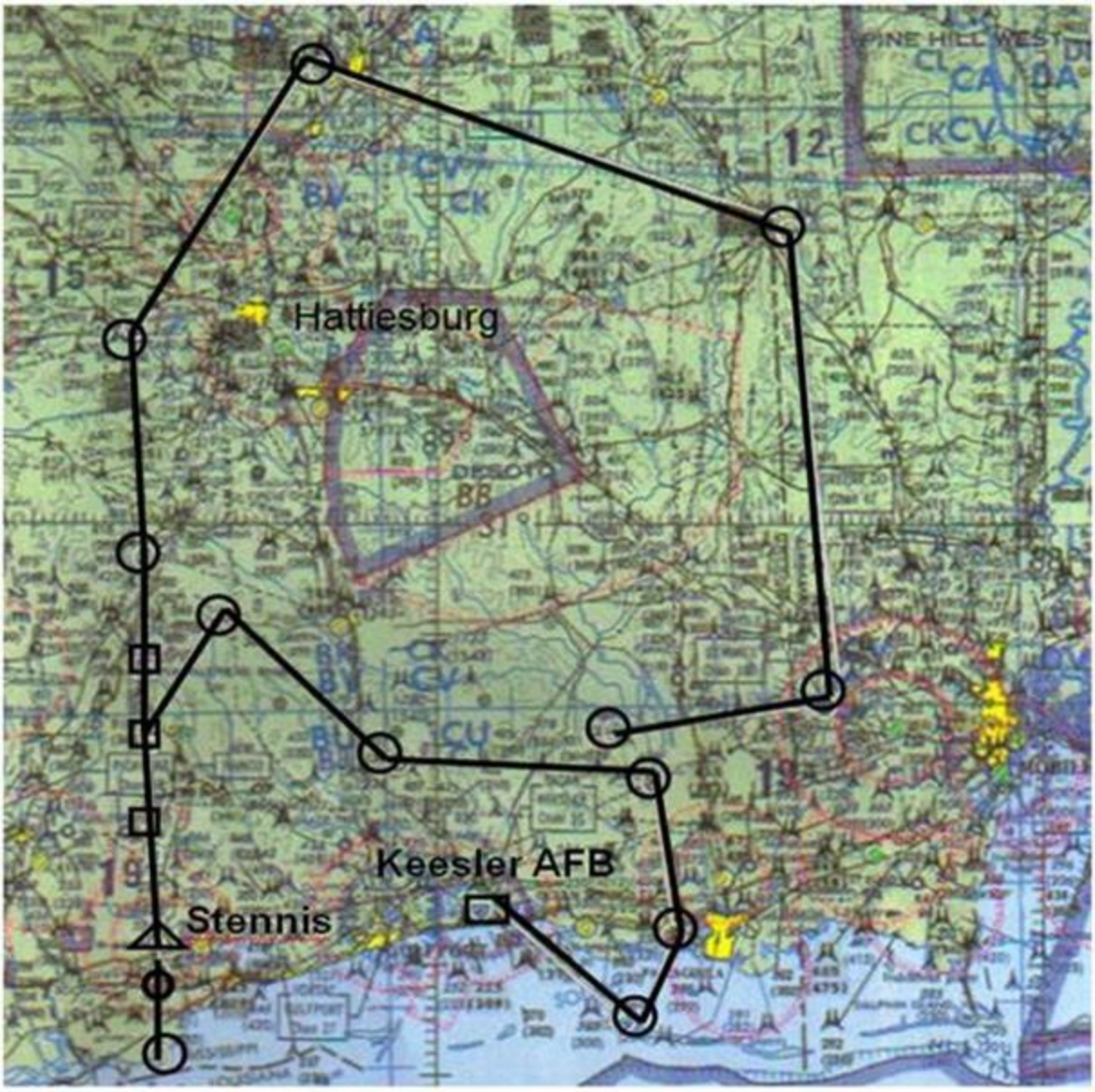


Figure A2.2. 53d Weather Reconnaissance Squadron.



Attachment 3

WEATHER IMPACT ON KEESLER AFB UNIT OPERATIONS

Table A3.1. 81FSS Food Services.

Weather Condition	Impact	Customer Action
Tornadic Activity	Causes delays in troop movement.	Any delay in troop movement causing the dining facilities to remain open longer than normal operating hours would increase the price of the contract.
Winds ≥ 35 kts	Causes delays in troop movement.	Any delay in troop movement causing the dining facilities to remain open longer than normal operating hours would increase the price of the contract.
Hail ≥ 1/2 inch	Causes delays in troop movement.	Any delay in troop movement causing the dining facilities to remain open longer than normal operating hours would increase the price of the contract.

Table A3.2. 81FSS Fitness Center.

Weather Condition	Impact	Customer Action
Tornadic Activity	All indoor/outdoor activities will cease.	Staff and troops will be sheltered until “All Clear” is given.
Winds ≥ 35 kts	Outdoor events will cease and equipment items will have to be stored inside.	Customers will discontinue outdoor sporting and fitness activities.
Hail ≥ 1/2 inch	Government vehicles will need to be sheltered. All outdoor activities will cease.	Customers will discontinue outdoor sporting and fitness activities.
Freezing Temperatures (Temp ≤ 32°F for 2 hours or more)	Outdoor sporting events will cease.	Customers will discontinue outdoor sporting activities.

Table A3.3. 81FSS Outdoor Recreation.

Weather Condition	Impact	Customer Action
Tornadic Activity	Shuts down all operation within ODR.	All watercraft will be advised to seek shelter, pool personnel will secure operation and advise swimmers to stay in pool house, Fam Camp users will be advised to seek shelter in comfort stations.
Winds ≥35 kts	No water craft will be allowed out, pools will remain open unless other weather warnings issued.	Customers in outside activities will be advised to see shelter inside a building.
Freezing Temperatures (Temp ≤ 32°F for 2 hours or more)	Will cause exposed pipes to the outside pavilions and public fishing pier.	Customers again will be required to provide adequate clothing in the use of equipment and facilities.

Table A3.4. 81FSS Craft Shops.

Weather Condition	Impact	Customer Action
Tornadic Activity	Potential danger to life and property.	Shut down all computers and computer operated equipment, which stops work in several areas; depending on warning or watch notice, cover or move items and equipment away from windows. Notice will also determine evacuation/shelter procedures.
Lightning within 5NM (Observed)	Potential danger to life and property.	All computers shut down, monitor kiln operation, prepare flashlights.

Table A3.5. 81FSS Auto Hobby Shop.

Weather Condition	Impact	Customer Action
Tornadic Activity	Potential danger to life and property.	All outside work stops, oil storage tanks are secured, all portable tools, tool boxes, machines, trash cans etc. are secured or brought inside, Depending on warning or watch notice limit use of equipment, tools and lifts inside. Notice will also determine evacuation/shelter procedures.
Lightning within 5NM (Observed)	Potential danger to life and property.	All outside work stops and prepare flashlights.
Hail $\geq$ 1/2 inch	Potential danger to life and property.	All outside work stops, oil storage tanks are secured.
Winds $\geq$ 35 kts	Potential danger to life and property.	Shop staff secures any items outside, which might cause a safety hazard, garage doors are closed, which limits indoor work.
Freezing Temperatures (Temp $\leq$ 32°F for 2 hours or more)	Water pipes freezing.	Water lines are bled, no outside work is done, and Car Wash use is regulated depending on duration of freezing weather.

Table A3.6. 81FSS Child Development and Youth Centers (Includes Family Child Care Homes).

Weather Condition	Impact	Customer Action
Tornadic Activity	Interrupts all outside activity and normal routines within classrooms. Emergency procedures implemented.	Staff, providers, and children will be sheltered within hallways, away from windows and any possible flying objects until "All Clear" is given. Outdoor activities stop.
Winds ≥ 35 kts	Outdoor events will cease and outdoor small equipment items will be secured.	Staff, providers, and children will discontinue outdoor activities.
Hail ≥ 1/2 inch	Potential danger to life and property.	All outside play and work stops.
Hot Temperatures		
(Yellow or Red Flag) WBGT is ≥ 85-89.9 degrees	*Outdoor activities limited*	Outdoor activities limited to 30 minutes periods until temperature drops below 85 degrees.
(Black Flag) WBGT is ≥ 90 degrees	*Outside activities suspended*	Children, staff, providers must remain inside until all clear given.
Cold Weather Conditions		
Wind chill factor is 49-40	*Outdoor play precautions taken.*	Jackets, sweaters should be worn for outdoor play.
Wind chill factor is < 40°F	*Limit outside activities.*	Limit outdoor play to 30 minutes periods or less.
Wind chill factor is < 32°F	*Outdoor activities suspended.*	Children, staff and children remain inside until wind chill is > 32°F.
Lightning within 5NM (Observed)	Follow guidance from Weather call.	All outdoor activities are stopped.

Table A3.7. 81TRW and 81TRG.

Weather Condition	Impact	Customer Action
Tornadic Activity	Potential danger to life and property. Causes delays in troop movements.	Troops will be sheltered until "All Clear" is given. Outdoor activity stops.
High Winds ≥ 50 kts	Causes delays in troop movements.	Students will require bussing. Outdoor activity stops.
Hail ≥ 1/2 inch	Causes delays in troop movements.	Students will require bussing. Outdoor activity stops.
Lightning within 5NM (Observed)	Causes delays in troop movements.	Students will require bussing. Outdoor activity stops.
Heavy Rain ≥ 2 inches in 2 hours	Causes delays in troop movements.	Students will require bussing. Outdoor activity stops.
Heavy Rain ≥ 6 inches in 12 hours	Causes delays in troop movements.	Students will require bussing. Outdoor activity stops. Possible cancellation of classes.
Heavy Snow ≥ 2 inches in 2 hours	Causes delays in troop movements.	Students will require bussing. Outdoor activity stops.
Freezing Precipitation	Causes delays in troop movements.	Students will require bussing. Outdoor activity stops.
Wind Chill -15°F or lower	Causes delays in troop movements.	Students will require bussing. Outdoor activity is curtailed.

Table A3.8. 81MDG.

Weather Condition	Impact	Customer Action
Tornadic Activity	Potential danger to life and property. All MDG patient services, access to the flight line, launch/recovery of regional air transportation mission, emergency vehicle response/operation ceases. Disasterresponse plans, Wing/MDG deployment commitments, Federal response Plan for NDMS patient reception/redistribution, Public Health ford inspections/cold storage, vector borne disease surveillance, Biomedical Services monitoring of base water/air/environment safety implemented.	As needed: Appropriate Emergency Management MDG/ICC Checklist and MDG notifications initiated; activate Medical Control Center/Casualty Management Teams/Contingency Response Teams, Recall Procedures; Emergency Vehicle Operations modified by threat/visibility; contact GPMRC to coordinate Air Evacuation (AE); Patient packaging considerations outdoors; In-house bed-down of MDG personnel for 24h manning, discharge patients, contact high-risk patients for in-house bed-down and activate shelters.
Moderate Winds $\geq$ 35 kts but < 50 kts	Same as above/depends upon severity and damage.	As needed: same as above.
High Winds $\geq$ 50 kts	Same as above/depends upon severity and damage.	(AE coordinated through GPMRC) AE Missions determined by wind direction, Aircraft Model/Pilot/Airfield Management decision process.
Hail $\geq$ 1/2 inch	Same as above/depends upon severity and damage.	Actions matched to need by MDG/CC who determines the level of response required.
Lightning within 5NM (Observed)	Same as above/depends upon severity and damage.	AE patients will not enplane or deplane.
Heavy Rain $\geq$ 2 inches in 2 hours	Same as above/depends upon severity and damage.	(AE coordinated through GPMRC) AE Missions determined by wind direction, standing water on runway, Aircraft Model/Pilot/Airfield Management decision process; Vehicle operations determined by visibility.

Table A3.9. 81MSG.

Weather Condition	Impact	Customer Action
Tornadic Activity	Potential danger to life and property. Deliveries aborted.	All outdoor activity stops and personnel seek shelter.
Moderate Winds ≥ 35 kts but < 50 kts	Curtails the mission. Deliveries delayed.	Discontinue/cancel all outdoor activity and seek shelter.
High Winds ≥ 50 kts	Curtails the mission. Deliveries aborted.	Discontinue/cancel all outdoor activity and seek shelter. Secure loose equipment.
Hail ≥ 1/2 inch	Curtails outside activities. Deliveries delayed.	Discontinue/cancel all outdoor activity and seek shelter.
Lightning within 5NM (Observed)	Curtails outside activities. Deliveries delayed.	Discontinue/cancel all outdoor activity and seek shelter.
Heavy Rain ≥ 2 inches in 2 hours	Curtails outside activities. Deliveries delayed.	Discontinue/cancel all outdoor activity. Secure doors in preparation for possible flooding.
Heavy Rain ≥ 6 inches in 12 hours	Curtails outside activities. Deliveries delayed.	Discontinue/cancel all outdoor activity. Secure doors in preparation for possible flooding.
Freezing Temperatures (Temp ≤ 32°F for 2 hours or more)	Possible cancellation of scheduled outdoor activities.	Warn outdoor fitness and sports participants of potential risk.
Freezing Precipitation	Curtails outside activities. Deliveries delayed.	Discontinue/cancel all outdoor activity.
Wind Chill -15°F or lower	Possible cancellation of scheduled outdoor activities.	Warn outdoor fitness and sports participants of potential risk. WBGT Flag system.
Heavy Rain > 6 inches in 12 hours	Same as above/depends upon severity and damage.	AE Missions determined by wind direction, standing water on runway, Aircraft Model/Pilot/Airfield Management decision process; Vehicle operations determined by visibility.

Freezing Temperatures (Temp < 32°F for 2 hours or more)	Vehicle operation/Pre- hospital patient packaging considerations.	Limited/curtailed pre-hospital emergency response, keep patients warm/dry, four-wheel drive/fire department escorts for manpower.
Temperatures below 40°F (Observed)	Vehicle operation/Pre- hospital patient packaging considerations.	Limited/curtailed pre-hospital emergency response, keep patients warm/dry.

Table A3.10. 85EIS.

Weather Condition	**Impact**	**Customer Action**
Tornadic Activity	Potential danger to life and property. Deliveries aborted.	All outdoor activity stops and personnel seek shelter.
Winds ≥ 50 kts	Stop all outside activity	All personnel take cover.
Winds ≥ 35 kts to < 50 kts	Stop most outside activity	Personnel in training cease activities and take cover, Utilize RM to determine if E & I activities are justified.
Hail ≥ 1/2 inch	Stop all outside activity	All Personnel take cover.
Heavy Rain or Snow (≥ 2 Inches in 2 Hours)	Curtail certain outside activity	Personnel in training cease activities and take cover, Utilize RM to determine if E & I activities are justified.
Heavy Rain (≥ 6 Inches in 12 Hours)	Curtail certain outside activity	Personnel in training cease activities and take cover, Utilize RM to determine if E & I activities are justified.
Freezing Precipitation	Curtail certain outside activity	Personnel in training cease activities and take cover, Utilize RM to determine if E & I activities are justified.
Blizzard	Curtail certain outside activity	Personnel take cover, Utilize RM to determine if E & I activities are justified.
Lightning within 5NM (Observed)	Stop all outside activity	All personnel take cover

Table A3.11. 81LRS.

Weather Condition	Impact	Customer Action
Tornadic Activity	Potential danger to life and property.	All outdoor activity stops and personnel seek shelter.
Lightning within 5NM (Observed)	Stops the mission.	All refueling is stopped. Personnel seek shelter.
Hail ≥1/2 inch	Possible delay in mission.	Personnel are warned and advised to take shelter as necessary.

Table A3.12. 81SFS.

Weather Condition	Impact	Customer Action
Tornadic Activity	Potential danger to life and property.	Patrols are warned to proceed with caution and to take shelter as necessary.
Hail ≥ 1/2 inch	Potential danger to life and property.	Patrols are warned to proceed with caution and to take shelter as necessary.
Lightning within 5NM (Observed)	Potential danger to life and property.	Patrols are warned to proceed with caution and to take shelter as necessary.
Heavy Rain ≥ 2 inches in 2 hours	Cause localized street flooding.	Extra patrols may be needed to put up barricades around flooded streets.
Heavy Rain ≥ 6 inches in 12 hours	Cause localized street flooding.	Extra patrols may be needed to put up barricades around flooded streets.
Heavy Snow ≥ 2 inches in 2 hours	Potential for increased vehicle accidents.	Extra patrols may be needed to respond to accidents.
Freezing Precipitation	Potential for increased vehicle accidents.	Extra patrols may be needed to respond to accidents.
Visibility < 1,500ft	Restricted visibility requires increased security.	Increase the number of patrols for base security.

Table A3.13. BOS/CE.

Weather Condition	Impact	Customer Action
Tornadic Activity	Potential danger to life and property.	All outdoor activity stops and personnel seek shelter.
Winds ≥ 25 kts	Curtails outside jobs. High altitude work stopped.	Discontinue/cancel work using bucket trucks, on roof tops or climbing pools.
Hail ≥ 1/2 inch	Curtails outside jobs.	Discontinue/cancel all outdoor work and seek shelter.
Lightning within 5NM (Observed)	Curtails outside jobs.	Discontinue/cancel all outdoor work and seek shelter.
Heavy Rain ≥ 2 inches in 2 hours	Curtails outside jobs.	1. Work using bucket trucks stops. 2. Sand bag low lying areas. 3. Excavation work is halted.
Heavy Rain ≥ 6 inches in 12 hours	Curtails outside jobs.	1. Work using bucket trucks stops. 2. Sand bag low lying areas. 3. Excavation work is halted.
Freezing Temperatures (Temp ≤ 32°F for 2 hours or more)	Possible cancellation of scheduled outdoor jobs.	1. Implement Freeze Protection Plan. 2. Boiler crews put on standby. 3. Place water pumps in manual mode.
Freezing Precipitation	Possible cancellation of scheduled outdoor jobs.	1. Implement Freeze Protection Plan. 2. Boiler and high voltage crews put on standby. 3. Place water pumps in manual mode.
Blizzard	Possible cancellation of scheduled outdoor jobs.	1. Implement Freeze Protection Plan. 2. Boiler and high voltage crews put on standby. 3. Place water pumps in manual mode.
Heavy Snow (≥ 2 Inches in 2 Hours)	Possible cancellation of scheduled outdoor jobs.	1. Implement Freeze Protection Plan. 2. Boiler crews put on standby. 3. Place water pumps in manual mode.

Table A3.14. 81CS Telephone Maintenance.

Weather Condition	Impact	Customer Action
Tornadic Activity	Possible destruction of communication facilities.	In case of total loss of telecommunications capability, request for Mobility Communication Unit for assistance.
Lightning within 5NM (Observed)	Technicians not able to respond to outages or work orders.	Due to close contact to electronic equipment and wiring, SCMP ceases all maintenance activities and takes cover.

Table A3.15. 81OSF/OSM and 81 CS Cable Shop and LMR Shop.

Weather Condition	Impact	Customer Action
Tornado	Stop all outside activity/maintenance.	Personnel take cover.
Winds $\geq$ 50 kts	Stop all outside activity/maintenance.	Personnel take cover.
Winds $\geq$ 35 kts to < 50 kts	Stop most outside activity/maintenance.	Personnel take cover, Utilize RM to determine if maintenance activities are justified.
Hail $\geq$ 1/2 inch	Stop all outside activity/maintenance.	Personnel take cover.
Heavy Rain or Snow ($\geq$ 2 Inches in 2 Hours)	Curtail certain outside activity/maintenance.	Personnel take cover, Utilize RM to determine if maintenance activities are justified.
Heavy Rain ($\geq$ 6 Inches in 12 Hours)	Curtail certain outside activity/maintenance.	Personnel take cover, Utilize RM to determine if maintenance activities are justified.
Freezing Precipitation	Curtail certain outside activity/maintenance.	Personnel take cover, Utilize RM to determine if maintenance activities are justified.
Blizzard	Curtail certain outside activity/maintenance.	Personnel take cover, Utilize RM to determine if maintenance activities are justified.
Lightning within 5NM (Observed)	Stop all outside activity. Technicians not able to respond to outages or work orders.	Due to close contact to electronic equipment and wiring, personnel cease all maintenance activities and take cover.

Table A3.16. 403WG MOC.

Weather Condition	Impact	Customer Action
Tornadic Activity	Potential danger to life and property.	All outdoor activity stops and personnel seek shelter.
Winds > 20 kts	Curtails outside maintenance.	1. AGE equipment not in use is stored. 2. Tie down aircraft nose landing gear. 3. Jacking terminates.
Winds > 30 kts.	Curtails outside maintenance.	1. Towing terminates. 2. Fueling terminates. 3. Loxing terminates. 4. Fuel cell repair terminates.
Winds > 50 kts	Outside maintenance stops.	1. Face aircraft into the wind. 2. Moore aircraft. 3. Remove AGE equipment from flight line. 4. Remove loose equipment and fire extinguishers.
High Winds ≥ 75 kts	Curtails outside maintenance.	1. Inspect surface controls and attachment points. 2. Evacuate aircraft to weather safe area.
Hail ≥ 1/2 inch	Potential danger to life and property.	1. Stops outside maintenance. 2. Hangar aircraft.
Lightning within 5NM (Observed)	Potential danger to life and property.	1. Stops outside maintenance.
Freezing Temperatures (Temp ≤ 32°F for 2 hours or more)	Limits outside maintenance.	Deice aircraft as necessary.
Freezing Precipitation	Limits outside maintenance.	Deice aircraft as necessary.
Temperatures below 40°F (Observed)	Wash rack will not wash aircraft.	Washer will wait until the temperature is above 40°F before washing aircraft.

Attachment 4

FLYING AREA MISSION EXECUTION FORECAST (FAMEF)

Figure A4.1. Tab1 of FAMEF.

KEESLER AFB FAMEF # 15-01									
DATE: February 15, 2024									
FAMEF USE FOR LOCAL FLYING ONLY (KBIX, KGPT, KMOB, KHSA, KJAN, KBTR, KMLU, KEFD, KNBG)									
FORECASTER'S INITIALS:	MAC	FAMEF ISSUED AT:	1345Z	FAMEF AMENDED AT:					
ICAO	VALID TIME (Z)			WINDS (KT)	VIS/WX	CLOUDS	ALSTG	TEMP	PA (FT)
KBIX	1545	TO	2000	12012	7	BKN035	3006	+15	-96
	2000	TO	0045	16012	5 VCSH BR	BKN010 OVC030	2999		-31
TEMPO	2000	TO	0045		3 SHRA	OVC005			
		TO							
KGPT	1545	TO	0045	13012	7	BKN030	3005	+16	-91
	1900	TO	0045	17012	5 VCSH BR	BKN010 OVC030	2999		-36
TEMPO	1900	TO	0045		3 SHRA	OVC005			
		TO							
KBHM	1545	TO	2100	11012	7	SCT050	3007	+14	81
	2100	TO	0045	18009	7 VCSH	BKN010 OVC025	3001		136
TEMPO	2200	TO	0045		5 -SHRA	OVC008			
		TO							
KHSA	1545	TO	1900	13012	7	BKN025	3007	+16	-115
	1900	TO	0045	17015	5 VCSH BR	BKN010 OVC025	3000		-50
TEMPO	2000	TO	0045		3 SHRA	OVC005			
		TO							
KJAN	1545	TO	1700	16009	7	BKN025	3002	+12	254
	1700	TO	0045	17010G20	7 VCSH	BKN009	2992		346
TEMPO	1900	TO	0045		4 -TSRA	BKN005 OVC010CB			
		TO							
KBTR	1545	TO	1700	14009	7	SCT010 BKN025	3005	+17	-49
	1700	TO	2100	16012	4 -SHRA BR	BKN008 OVC015	2998		15
	2100	TO	0045	17009	6 BR	BKN008 OVC015	2994		52
		TO							
Alternates (only filled in when primary airfields are <2000ft/3mi) or when required									
KMLU	1545	TO	2000	18010G18	7	OVC007	3004	+18	-31
	2000	TO	0045	20015G25	7	OVC007	2995		51
		TO							
KEFD	1545	TO	1800	VRB06	3 BR	OVC005	2999	+18	-31
TEMPO	1545	TO	1800		1/2 FG	OVC003			
	1800	TO	0045	26009	6 BR	OVC010	2988		70
KNBG	1500	TO	1900	18010	7	BKN015 OVC030	3006	+19	-127
	1900	TO	0045	20009	5 TSRA BR	BKN015CB OVC030	2999		-62
TEMPO	2100	TO	0045		1 SHRA	BKN005 OVC010			

MINIMUM CEILING - LOCATION	MAXIMUM CLOUD TOPS - LOCATION	MINIMUM FREEZING LVL - LOCATION
005 - ENRTE	250 - N GULF	020 - Cent MS

FLT LEVEL WINDS — CLICK HERE FOR OTHER FLIGHT LEVEL WINDS

LEVEL	010	030	050	100	140	180
SPEED (KT)	18020	22030	23035	24040	26055	27060

HAZARDS (CLICK ON LINK IF BOX IS RED)					SPACE WX		
LL TURB	UL TURB	LL ICING	UL ICING	TSTMS	HF	UHF	GPS

TODAY'S SOLAR DATA (LOCAL TIME)			TODAY'S LUNAR DATA (LOCAL TIME)		
SUNSET	EECT	EENT	MOONRISE	MOONSET	% ILLUMINATION
1744	1808	1836	1016	****	35

OTHER BRIEFING TOOLS			
IR METSAT	WV METSAT	VIS METSAT	OWS RADAR
OWS LTG	AFW-WEBS	ADDS AIRMETS\SIGMETS	ADDS PIREPS

IF YOU HAVE ANY QUESTION, CONTACT WX FLIGHT AT 377-3305 OR PMSV 267.4

Figure A4.2. Tab2 of FAMEF 815AS Drop Zone Mission Execution Forecast on Days with Drop Zone Missions.

815th AS - DROP ZONE FORECASTS
SEE FRONT PAGE FOR FORECASTS

BOFA DROP ZONE (KBIX)	
VALID TIMES:	
1630Z-1830Z	
FL WINDS(KT)/TEMP(C):	
SFC:	10009/+09
005:	13012/+10
010:	16015/+11
015:	17020/+10
020:	19020/+10
030:	22025/+10

RALFFIE DROP ZONE (KSH1)	
VALID TIMES:	
1630Z-1830Z	
FL WINDS(KT)/TEMP(C):	
SFC:	11010/+09
005:	12015/+09
010:	15015/+10
015:	17020/+10
020:	19020/+10
030:	22025/+10

TO BE FILLED FOR ASSAULT LANDINGS ONLY (WEDNESDAY)

ICAO	VALID TIME (Z)			WINDS	VIS/WX	SKY CONDITION	ALSTG	TEMP	PA (FT)
KSH1	1630Z	TO	1830Z	11010	7	BKN025	3002	+12	213
		TO							
		TO							

Figure A4.3. Tab3 of FAMEF 53rd Weather Reconnaissance Squadron - SFMR Calibration GULF OF MEXICO when flying SFMR Mission.

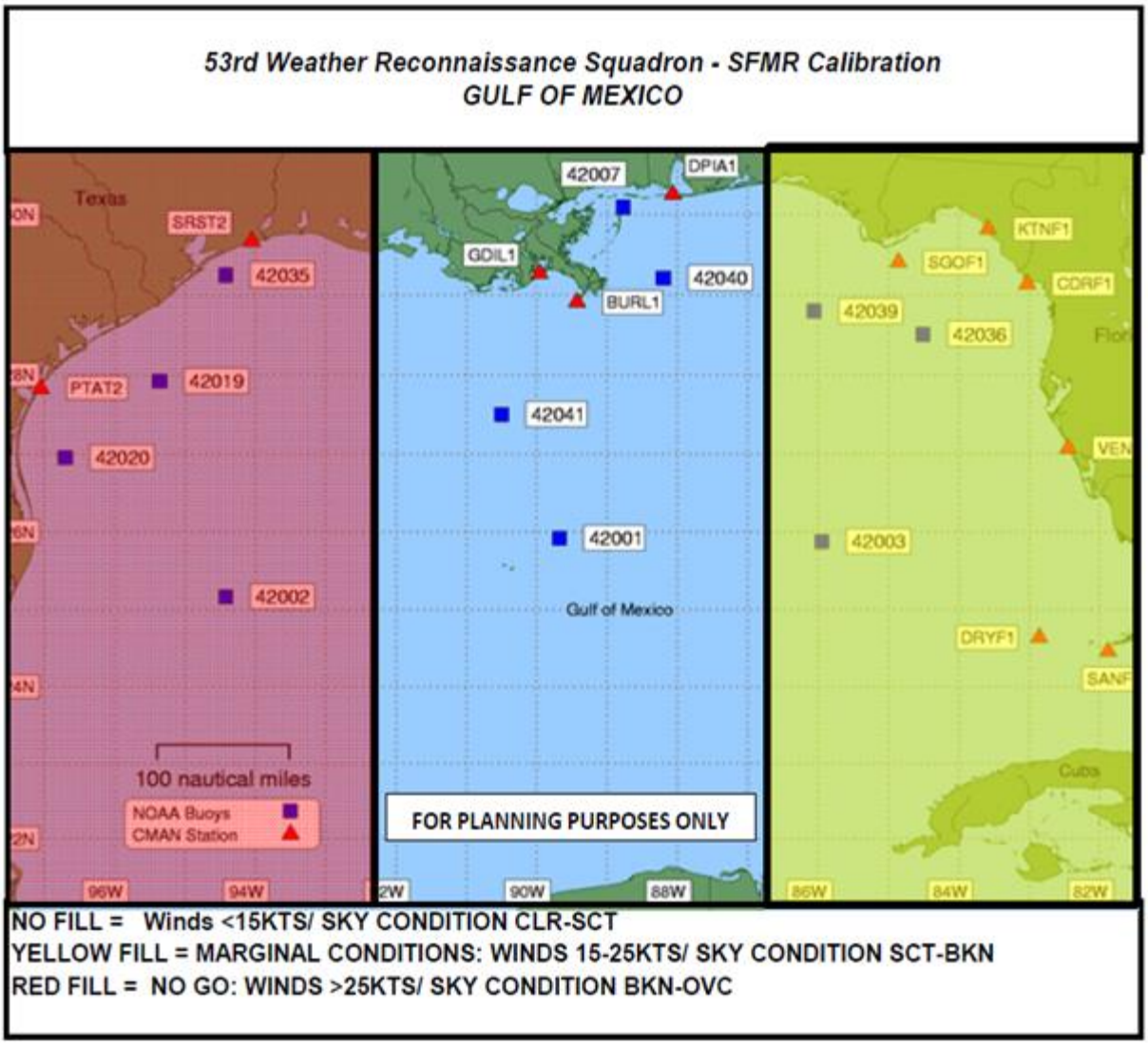


Figure A4.4. Tab4 of FAMEF 815AS Drop Zone Mission Execution Forecast for Steinhawk DZ.

STEINHAWK (KBIX)		
VALID TIMES: 1630Z-1830Z		
	FL Winds (KT):	Temps (C):
SFC:	12012	+15
005:	15010	+14
010:	18010	+12
015:	20010	+10
020:	23010	+10
030:	26015	+10
040:	28010	+08
050:	28007	+09
060:	29007	+08
070:	29008	+07
080:	30013	+06
090:	30015	+05
100:	29020	+03
110:	30020	+03
120:	28025	-02
130:	27035	-04

Attachment 5

DD FORM 175-1, FLIGHT WEATHER BRIEFING

Figure A5.1. DD Form 175-1 Flight Weather Briefing.

HELP IS ON | CLEAR ALL

FLIGHT WEATHER BRIEFING

CONVERSION CALCULATOR

PART I - TAKE OFF DATA (CLEAR PART I) (PART I HELP)

1. DATE(Z)	2. ACFT TYPE / NO	3. DEP PT / ETD	4. TEMP	5. DEW PT	6. TMP DEV	ALSTG (INS)	7. PRES ALT	8. DENSITY ALT
02 Feb 24	C130/ HOBBY09	KBIX / 2000 Z	+18 °C	+12 °C		29.92	+37 FT	+399 FT

9. SFC WIND (KT)	10. CLIMB WINDS	11. LOCAL WEATHER WATCH/WARNING/ADVISORY	12. RSC / RCR	
17009 M	NONE	NONE	DRY	N/A

13. REMARKS / TAKEOFF ALTN FCST

NONE

PART II - ENROUTE & MISSION DATA (CLEAR PART II) (PART II HELP)

14. FLIGHT LEVEL WINDS / TEMP°C — SEE ATTACHED

240	34055KT/-30C GA-NC
	28035KT/-28C MS-AR

15. SPACE WEATHER

	CLEAR	NO IMPACT	MARGINAL	SEVERE
HF		X		
UHF		X		
GPS		X		

SOLAR/LUNAR ? LOCATION VIEW CHART

SUNT	1202 Z	RATON RIDGE METRO	
SR	1256 Z	MR	0600 Z
SS	2345 Z	MS	1704 Z
EENT	0035 Z	ILLUM	51 %

17. CLOUDS AT FLT LEVEL — YES | NO | X IN AND OUT

18. OBSCURATIONS AT FL RESTRICTING VISIBILITY (OTHER THAN CLOUDS) — YES | X NO

19. MINIMUM CEILING - LOCATION: 015 - W NC FT AGL

20. MAXIMM CLOUD TOPS - LOCATION: 280 - NC FT MSL

21. MINIMUM FREEZING LEVEL - LOCATION: 040 - NC FT MSL

22. THUNDERSTORMS XXXX	23. TURBULENCE XXXX	24. ICING XXXX	25. PRECIPITATION XXXX
CHART 1WXG 0218Z/0306Z	CHART 1WXG 0218Z/0306Z	CHART 1WXG 0218Z/0306Z	CHART FCST
X NONE / AREA / LINE	NONE / IN CLEAR / IN CLOUD	NONE / RIME / MXD / CLR	X NONE / DRIZZLE / RAIN / SNOW / SLEET
ISOLATED 1-2%	LIGHT X	TRACE	LIGHT
FEW 1-3%	MODERATE	LIGHT X	MODERATE
SCATTERED 16-45%	SEVERE	MODERATE	HEAVY
NUMEROUS > 45%	EXTREME	SEVERE	SHOWERS
HAIL, SEVERE TURBULENCE & ICING, HEAVY PRECIPITATION, LIGHTNING & WIND SHEAR EXPECTED IN AND NEAR THUNDERSTORMS	LEVELS SFC-060	LEVELS 040-080	FREEZING
LOCATION ENRTE	LOCATION NC	LOCATION NC	LOCATION ENRTE

PART III - AERODROME FORECASTS (CLEAR PART III) (OBS & TAFS) (PART III HELP)

26. STATION	27. VALID TIME	28. SFC WIND (KTS)	29. VSBY/WX	30. CLOUD LAYERS	31. TEMP	32. ALSTG (INS)	33. PA
DESTINATION KLRF	2015 Z TO 2215 Z	VRB06 T	7SM	SCT030	+19 °C +66 °F	29.88	+356 FT
DESTINATION KPOB	2215 Z TO 0015 Z	33012 T	7SM	SCT040	+15 °C +59 °F	29.82	+294 FT
DESTINATION KFTY	2345 Z TO 0145 Z	32009 T	7SM	SKC	+13 °C +55 °F	29.93	+855 FT
DESTINATION KBIX	0115 Z TO 0315 Z	17009 T	7SM	FEW200	+12 °C +54 °F	29.94	+19 FT
DESTINATION KBTR	0215 Z TO 0415 Z	VRB06 T	7SM	SKC	+14 °C +57 °F	29.87	+120 FT
DEST / ALTN	TO	T			°C °F		FT
DEST / ALTN	TO	T			°C °F		FT
DEST / ALTN	TO	T			°C °F		FT
DEST / ALTN	TO	T			°C °F		FT
DEST / ALTN	TO	T			°C °F		FT

PART IV - COMMENTS/REMARKS

32. BRIEFED RSC/RCR — YES | X NOT AVAILABLE — KBIX 33. PMSV 267.4 — 34. ATTACHMENTS — YES | X NO

35. REMARKS

"WEATHER BRIEFINGS PROVIDED ELECTRONICALLY (E.G., FAXED, POSTED ON WEB PAGE, OR E-MAILED) ARE REQUIRED TO CALL KEESLER AFB AT DSN 597-3305 OR COMMERCIAL (228) 377-3305 FOR A WEATHER UPDATE."

PART V - BRIEFING RECORD (CLEAR PART IV & V)

36. WX BRIEFED TIME	37. FLIMSY BRIEFING NO	38. FORECASTER'S INITIALS	39. NAME OF PERSON RECEIVING BRIEFING
E02/1800Z //		MAC	LTC LAMBERT
40. VOID TIME	41. EXTENDED TO / INITIALS	42. WX REBRIEF TIME / INITIALS	43. WX DEBRIEF TIME / INITIALS

Figure A5.2. Back Page of DD Form 175-1 Follow on Support.

FOLLOW ON WEATHER SUPPORT

Obtain weather support at next location from the agency indicated below.
REQUEST BRIEFING 2 HOURS IN ADVANCE OF EXPECTED BRIEF TIME.
1: Primary Support 2: Secondary support 3: Third supporting agency

1 Keesler AFB: **DSN 597-3305** Com (228) 377-3305 **Briefing Hours: 0700L - 2300L / 1000L-1700L (Week-ends)**

2 26 OWS-Barksdale AFB, LA: **DSN 331-2651** Com (318) 529-2651

_____ 25 OWS-Davis Monthan AFB, AZ: **DSN 228-6598** Com 1-877-451-8367

_____ 28 OWS-SHAW AFB, SC: DSN 965-0906 COM (803)895-0906 (FOR SOUTHWEST ASIA SUPPORT ONLY)

_____ 15 OWS-Scott AFB, IL: **DSN 576-9701** Com (618) 256-9701

_____ Other: _____

Attachment 6

KEESLER AFB HURRICANE BULLETIN

Figure A6.1. Keesler AFB Hurricane Bulletin.

**********FOR NON-OPERATIONAL USE ONLY**********

Keesler AFB Hurricane Bulletin

Hurricane Advisory #5 (NHC #17) -- Date: 27 SEP 2022 -- Time: 0400L CDT



Legend: Green Ring = 34 knots sustained; Blue Ring = 50 knots sustained; Red Ring = 64 knots sustained; Dark Red Ring = Center. Grey shaded area is the Forecast Error Area (where the storm could go besides the forecasted track).
Forecast track (red line) extends up to 120 hours depending on map scale and includes date/time and center wind speed.
Errors for track have averaged near 225NM on DAY 4 and 300NM on DAY 5 and for intensity by 20 knots each day.

Current Information: HURRICANE IAN
Position: 22.3N 83.7W → ···· → **Movement:** N@10kts (12mph) → → **Current Category:** CAT 3
Sustained Wind: 110kts (125mph) → → **Gusts:** 135kts (155mph)
Location from KAFB: 560 NM SE of KAFB → → **CPA of KAFB:** 345 NM E of KAFB
Watches/Warnings: Hurricane Warning in effect for Dry Tortugas, Cuban provinces, western FL coast; Trop Storm Warning for southwestern FL coast, FL Keys, and Cuban provinces; Hurricane Watch for northwestern and southwestern FL coast; Trop Storm Watch for eastern FL coast, and northern FL coast extending west to Apalachicola.

Storm Discussion: Hurricane Ian is a major category 3 storm moving north over Western with continued strengthening forecast over the next 24hrs. Hurricane Ian is forecasted to continue moving north away from Cuba into the southeastern Gulf of Mexico later today. Hurricane Ian will move past the Keys to the west late Tuesday and begin to approach the west coast of Florida on Wednesday. A slow turn to the north-northeast is forecasted to occur on Wednesday with slight weakening before making landfall on Thursday.
Effects on Keesler AFB: NONE
Expected Landfall: Near Tampa, FL Thursday morning

OTHER USEFUL INFORMATION: 1 knot (kt) = 1.15 miles per hour (mph); 1 nautical mile (nm) = 1.15 statute miles (mi)

HURCON/TCCOR (Hurricane/Tropical Cyclone Conditions)	→	Hurricane Categories:
5 = Destructive winds are **possible** within 96 hours	→	1 = sustained winds of 64-82 knots (74-95 mph)
4 = Destructive winds are **possible** within 72 hours	→	2 = sustained winds of 83-95 knots (96-110 mph)
3 = Destructive winds are **probable** within 48 hours	→	3 = sustained winds of 96-112 knots (111-129 mph)
2 = Destructive winds are **expected** within 24 hours	→	4 = sustained winds of 113-136 knots (130-156 mph)
1 = Destructive winds are **imminent** within 12 hours	→	5 = sustained winds 137 knots (157 mph) or greater

Created By: BOS/WEATHER ····· **For Operational Needs Only, Call:** 377-3305. ····· **Issued by: SB**

**********FOR NON-OPERATIONAL USE ONLY**********

Attachment 7

KEESLER AFB 5-DAY FORECAST

Figure A7.1. Slide two of Keesler AFB 5-Day Forecast.

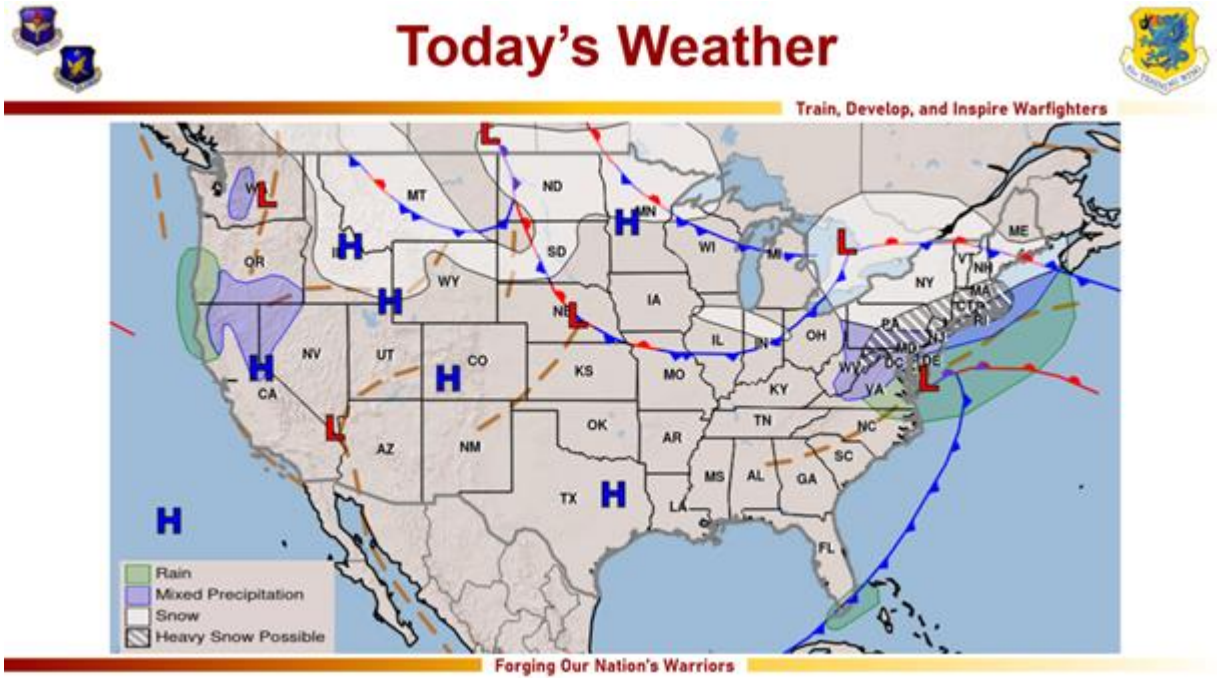


Figure A7.2. Slide three of Keesler AFB 5-Day Forecast.

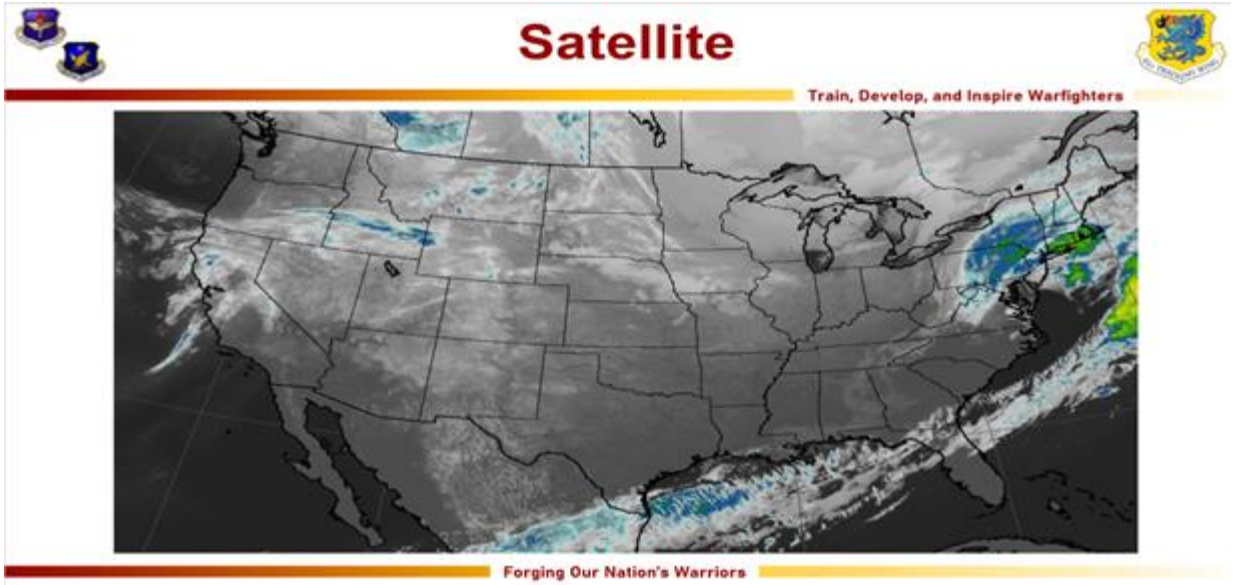


Figure A7.3. Slide four of Keesler AFB 5-Day Forecast.

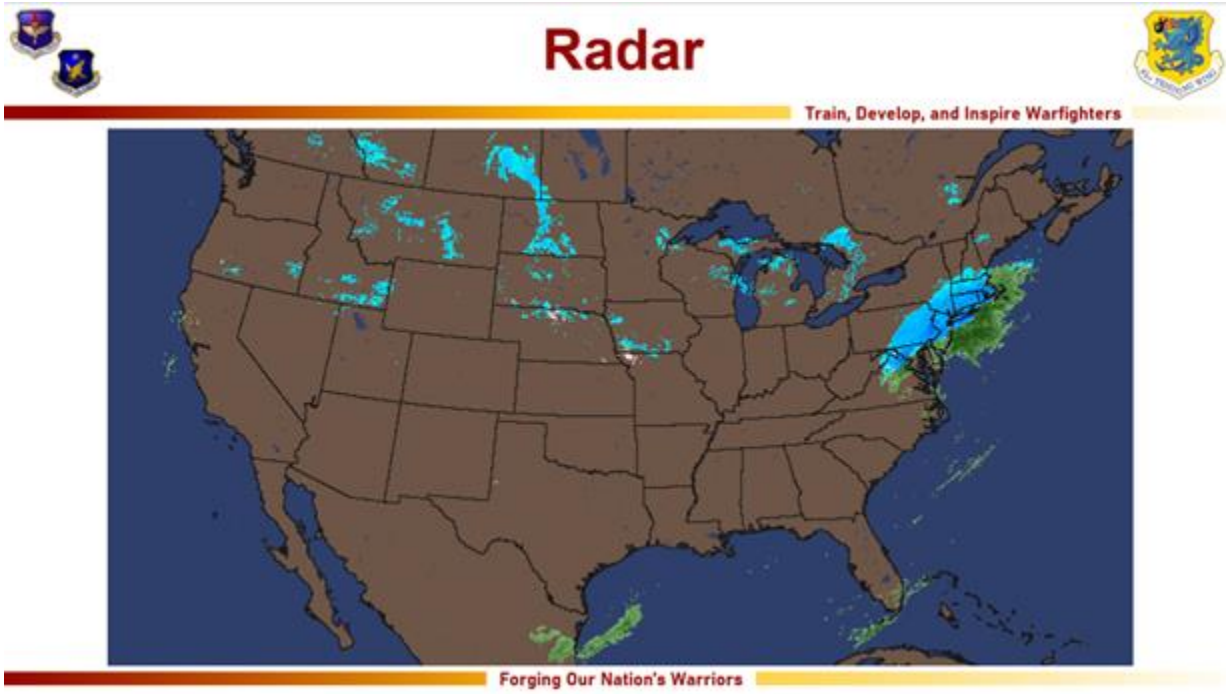


Figure A7.4. Slide five of Keesler AFB 5-Day Forecast.

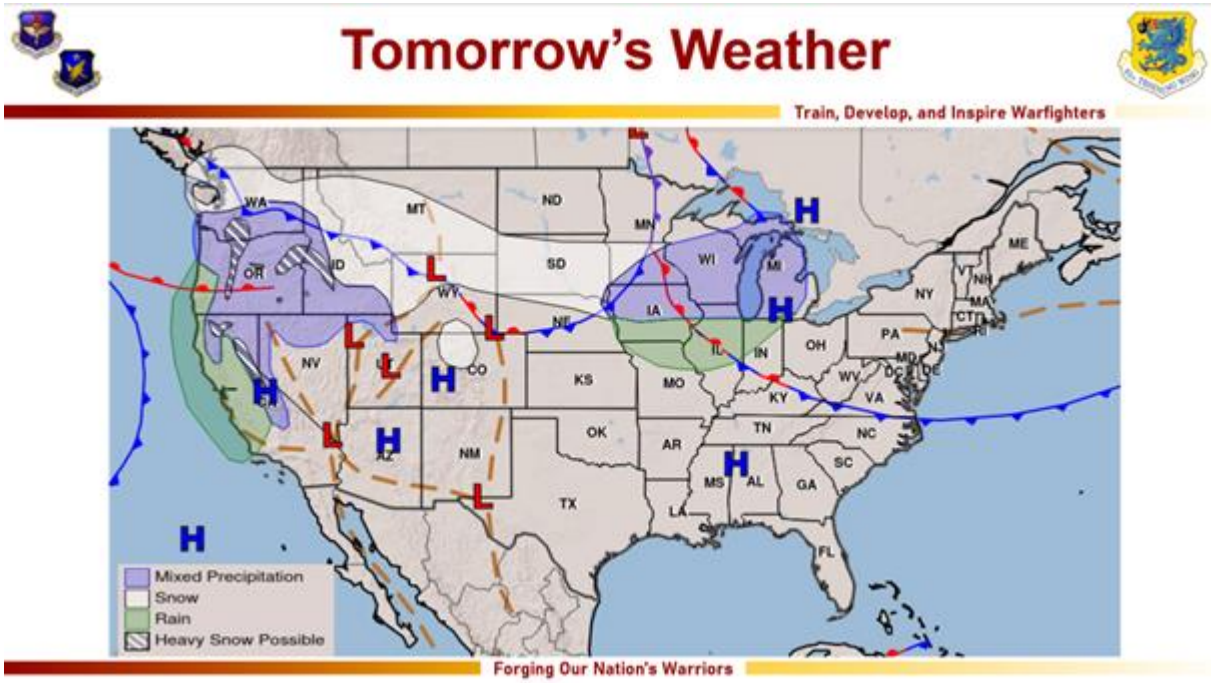


Figure A7.5. Slide six of Keesler AFB 5-Day Forecast.

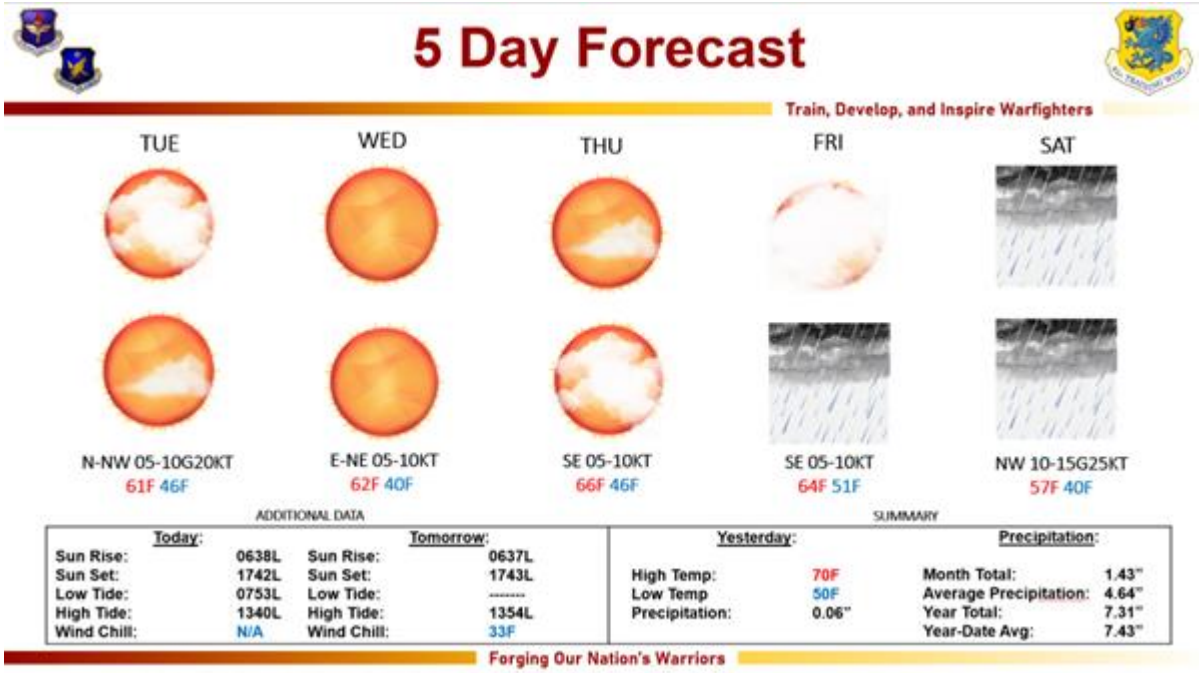
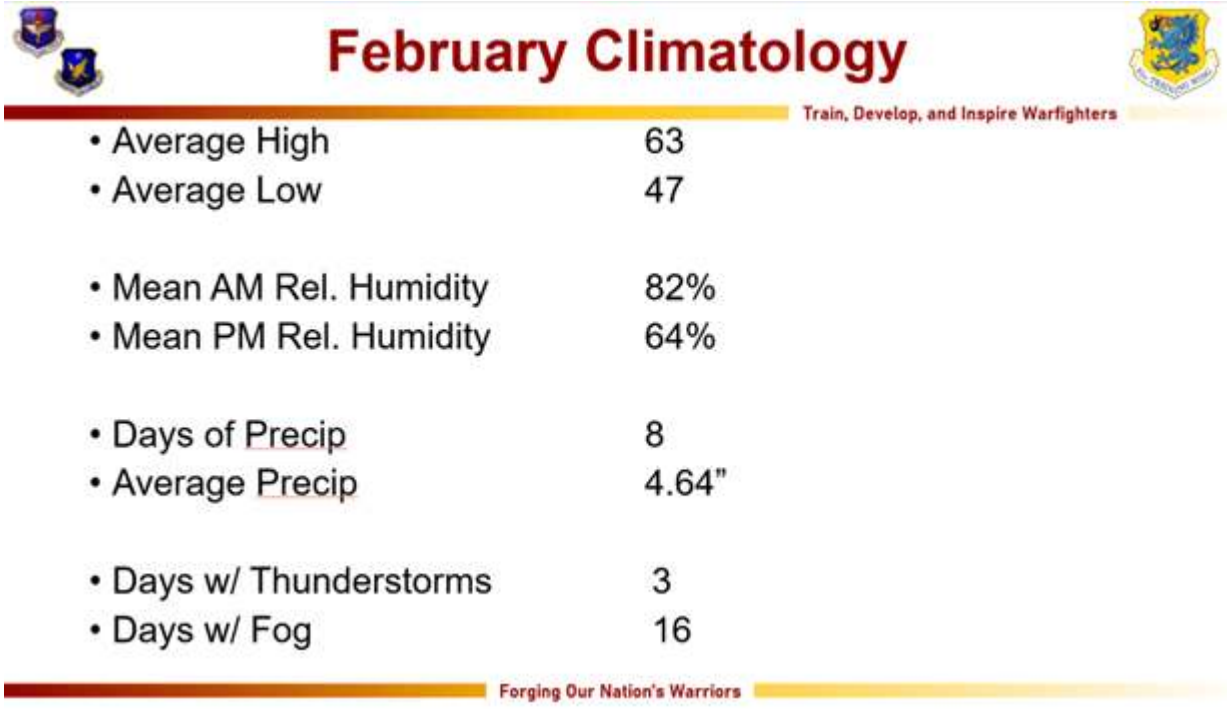


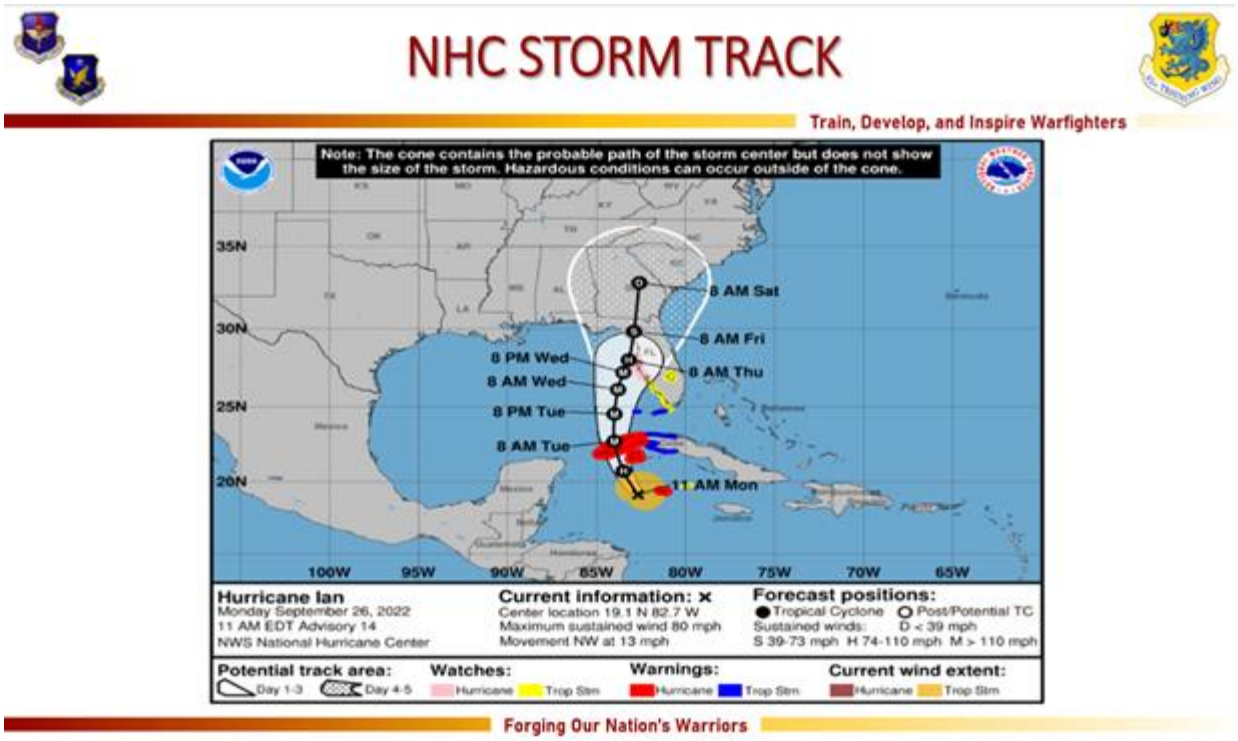
Figure A7.6. Slide seven of Keesler AFB 5-Day Forecast.



Attachment 8

KEESLER AFB TROPICAL SLIDES

Figure A8.1. Added to Keesler AFB 5-Day Forecast during Hurricane Season.



Attachment 9

FORECAST SPECIFICATION AND AMENDMENT CRITERIA

A9.1. General. TAFs are forecasts of meteorological elements expected to occur at a specific terminal. They are routinely prepared and are used for operational purposes. Specified in each TAF (scheduled and amended) is the expected time of occurrence to the nearest hour (and minute as appropriate), the expected duration, and intensity (where applicable) when certain conditions are expected to occur.

A9.1.1. Terms explained:

A9.1.1.1. Terminal Aerodrome. The area within a 5SM radius of the KEESLER AFB runway complex.

A9.1.1.2. Terminal Aerodrome Forecast. This is a forecast of required weather elements for a particular terminal aerodrome covering a 30-hour period. The TAF is a concise statement of the expected meteorological conditions at the terminal aerodrome during the specified period.

A9.1.1.3. Vicinity. The area between 5SM and 10SM radius of the KEESLER AFB runway complex.

A9.1.1.4. BECMG**.** The change indicator for the TAF. A change group of "BECMG 1719" becomes valid at 19Z; "BECMG 1213" is valid at 13Z, etc.

A9.1.1.5. TEMPO. This term indicates intermittent weather conditions for the TAF. A grouping of TEMPO 1519 indicates intermittent conditions will occur from 15Z-19Z.

A9.1.1.6. FM. Is used to forecast a change expected to take place quickly over a period of less than 30 minutes. "FM13" becomes valid at 1300Z and expects the specified condition to take place by 1330Z.

A9.1.1.7. Amendment. An unscheduled forecast which revises the current forecast.

A9.2. Amendments. TAFs may be amended at any time to ensure they are representative. Amendments are valid for the period starting the whole hour prior to the amendment and cover the remaining time of the original forecast.

A9.2.1. The TAF will be amended when:

A9.2.1.1. An unforecasted change which meets amendment criteria is expected to occur, or occurs, and is expected to last at least 30 minutes and is not correctly forecast by the next whole hour.

A9.2.1.2. A forecast condition meeting amendment criteria does not occur by the specified hour and is not expected to occur within the next 30 minutes.

A9.2.1.3. When a TEMPO group becomes predominant or is not expected to occur.

A9.2.1.4. Anytime in the interest of flight safety.

A9.2.2. Amendment Criteria:

Table A9.1. Keesler AFB Terminal Aerodrome Forecast Specification and Amendment Criteria.

Forecast Element/Occurrence	**TAF Amendment Criteria**	
Ceiling or Visibility observed or later expected to decrease to less than, or if below, increase to equal or exceed:	**Category**	**Limits**
	E	≥ 2,000 feet/4,800 meters (3 statute miles)
	D	2,000/4,800 and ≥ 1,000 feet/3,200 Meters (2 statute miles)
	C	< 1,000/3,200 and ≥ 700/0800 meters (1/2 statute miles)
	B	700 and ≥ 200
	A	<200/0800 meters (1/2 statute mile)
	Categories are determined by the lower of the values. Notes: 1. Forecast category is determined by the lower ceiling or visibility value. **2.** Use surface visibility to determine forecast category. **3.** Substitute the lowest published airfield minimum for Ceiling/Visibility category A. (may require deletion of the B category at some airfields).	
Surface Winds:	Wind Speed: The difference between the predominant wind speed and the forecast wind speed is ≥ 10 knots and/or the difference between the observed gusts is ≥ 10 knots from the forecast gust. For example, amend a forecast specifying surface winds of 23018G25KT if observed predominant wind speed is 28 knots or more, or if the observed gusts are 35 knots or higher. Similarly, amend the TAF if predominate winds are 8 knots or less, or gusts are 15 knots or less. Wind Direction: A change > 30 degrees when the predominant wind speed or gusts are expected to be over 15 knots.	
Icing not associated with thunderstorms, from the surface to 10,000 feet AGL.	The beginning or ending of icing first meets, exceeds, or decreases below moderate or greater thresholds and was not specified in the forecast.	
Turbulence (for Cat II aircraft), not associated with thunderstorms, from the surface to 10,000 feet AGL.	The beginning or ending of turbulence first meets, exceeds, or decreases below moderate or greater thresholds and was not specified in the forecast.	

Forecast **WW** criteria and/or TAF Amendable **WA** criteria – including non-convective low-level winds shear: **Note*:*** Watches or Advisories issued for an area may or may not be specified in a TAF coded forecast for an installation depending on situation.	Occur, or are expected to occur, during the forecast period, but were not specified in the forecast. Were specified in the forecast, but is no longer occurring or expected to occur during the forecast period.
Thunderstorms:	Incorrect forecast start or end time.
Specification of Temporary Conditions:	Forecast conditions specified as temporary become predominant conditions. Forecast conditions specified as temporary do not occur as forecast. Forecast conditions specified as temporary are no longer expected to occur.
Changes to Predominant Conditions:	Forecast change conditions (BECMG or FM group) occur before the beginning of the specified period of change and are expected to persist. Forecast change conditions (BECMG or FM group) do not occur by the specified time. Forecast change conditions (BECMG or FM group) are no longer expected to occur.
Representative Conditions:	Forecast conditions are considered unrepresentative of existing or forecast conditions and amending the forecast improves safety, flight planning, operations efficiency or assistance to in-flight aircraft.

Attachment 10

SAMPLE DISSEMINATION FORMATS AND INTERPRETATION

A10.1. Keesler AFB header and watch, warning or advisory number.

A10.1.1. Watches, warnings and advisories will have a separate numbering system and will be formatted as MM-NNN. The first two digits (MM) will signify the month, and the last 3 digits (NNN) will signify the watch, warning or advisory number. Example: The second advisory for the month of November would be 11-002.

A10.1.2. It is possible to have a watch, warning or advisory out at the same time with the same number. The key in determining the difference is the header. See the headers in A10.2.1., A10.3.1. A10.3.2., and A10.4.1. for examples.

A10.1.3. It is possible to have more than one watch, warning and advisory out at one time. It is important that our customers read each product carefully.

A10.1.4. Valid times on all watches, warnings and advisories will be in UTC and local time.

A10.2. Weather Watch Example. Here is an example of a weather watch:

A10.2.1. KEESLER AFB WEATHER WATCH 09-002

A10.2.1.1. VALID 05/1100Z (05/0600L) TO 05/1500Z (05/1000L)

A10.2.1.2. THE POTENTIAL EXISTS FOR SEVERE THUNDERSTORMS AT KEESLER AFB. SEVERE STORMS CAN PRODUCE WINDS $\geq$ 50 KNOTS AND HAIL $\geq$ 3/4 INCH DIAMETER. A WARNING WILL BE ISSUED IF CONDITIONS WARRANT.

A10.3. Weather Warning Example. Here are examples of a forecast weather warning and an observed weather warning, respectively:

A10.3.1. KEESLER AFB WEATHER WARNING 09-001

A10.3.1.1. VALID 05/1300Z (05/0800L) TO 05/1400Z (05/0900L)

A10.3.1.2. SEVERE STORMS ARE EXPECTED AT KEESLER AFB, ALONG WITH WINDS OF 50 KNOTS AND HAIL OF 3/4 INCH.

A10.3.2. KEESLER AFB WEATHER WARNING 10-004

A10.3.2.1. VALID 19/1921Z (19/1421L) TO UFN (UFN)

A10.3.2.2. LIGHTNING IS OBSERVED WITHIN 5NM OF KEESLER AFB.

A10.4. Weather Advisory Example. Here is an example of weather advisory:

A10.4.1. KEESLER AFB WEATHER ADVISORY 09-024

A10.4.1.1. VALID 18/1714Z (18/1214L) TO 18/2300Z (18/1800l)SURFACE WINDS OF 25-34 KNOTS ARE EXPECTED AT KEESLER AFB. MAX WIND OF 30 KNOTS IS FORECASTED.

A10.5. TAF Format.

A10.5.1. - (1)(2)(3) (4) (5)(6)(7)

A10.5.1.1. KBIX FCST 2410/2516 06008KT 7 SCT050 SCT120 ALTIMETER30.10INS WND 17012KT AFT 2418Z (8) BECMG 2419/2420 18012KT 7 –RA SCT010 BKN020 OVC050 ALTIMETER30.00INS TEMP 27C AT 2420Z TEMP 20C AT 2411Z 18/RI (9)(10)

A10.5.1.2. 1 - Station identifier

A10.5.1.3. 2 - Valid time of the forecast (30 hour forecast period)

A10.5.1.4. 3 - Wind direction and speed

A10.5.1.5. 4 - Visibility and obstructions to vision (if any)

A10.5.1.6. 5 - Sky condition

A10.5.1.7. 6 - Lowest altimeter setting expected

A10.5.1.8. 7 - Remarks (if any)

A10.5.1.9. 8 - Time of the next forecast group

A10.5.1.10. 9 - Maximum and minimum temperature forecast during TAF with

A10.5.1.11. time of occurrence for each

A10.5.1.12. 10 - Minutes after the hour/initials

A10.6. Record Observation (METAR) Format (Automated Observation).

A10.6.1. - (1)(2)(3)(4)(5)(6)(7)(8)(9)

A10.6.1.1. KBIXMETAR0955Z03005KT1/2SMR21/2400FTFGSCT030 06/M02 A3038 RMK AO2 SLP287 T00621022 $ PA +7 DA +2263

A10.6.1.2. - (10)

A10.6.1.3. 1 - Station identifier and observation type

A10.6.1.4. 2 - Time

A10.6.1.5. 3 - Wind direction and speed

A10.6.1.6. 4 - Visibility

A10.6.1.7. 5 - Runway visual range (when required; values sent locally only)

A10.6.1.8. 6 - Weather and obstruction to vision (if any)

A10.6.1.9. 7 - Sky condition

A10.6.1.10. 8 - Temperature and dew point

A10.6.1.11. 9 - Altimeter setting

A10.6.1.12. 10 - Remarks (as required) **Note**: AO2 signifies that the observations was automated

A10.7. Special Observation (SPECI) Format (Automated Observation).

A10.7.1. - (1)(2)(3)(4)(5)(6)(7)(8)(9)

A10.7.1.1. KBIXSPECI1010Z18013KT1/4SMR04/1600FTFGBKN008 19/18A2970 RMK AO2 PRESFR SLP057 $ PA +7 DA +2263

A10.7.1.2. - (10)(11)

A10.7.1.3. 1 - Station identifier and observation type

A10.7.1.4. 2 - Time

A10.7.1.5. 3 - Wind direction and speed

A10.7.1.6. 4 - Visibility

A10.7.1.7. 5 - Runway visual range (when required; values sent locally only)

A10.7.1.8. 6 - Weather and obstruction to vision (if any)

A10.7.1.9. 7 - Sky condition

A10.7.1.10. 8 - Temperature and dew point

A10.7.1.11. 9 - Altimeter setting

A10.7.1.12. - Remarks (as required) **Note**: AO2 signifies that the observations was automated

A10.8. Record Observation (METAR) Format (Supplemented Observation).

A10.8.1. - (1)(2)(3)(4)(5)(6)(7)(8)(9)

A10.8.1.1. KBIXMETAR1255Z12003KT1/8SMR22/1200FTFGOVC002 20/19 A3010 RMK AO2A SLP287 T00621022 $ PA +7 DA +226357/BF

A10.8.1.2. - (10)(11)

A10.8.1.3. 1 - Station identifier and observation type

A10.8.1.4. 2 - Time

A10.8.1.5. 3 - Wind direction and speed

A10.8.1.6. 4 - Visibility

A10.8.1.7. 5 - Runway visual range (when required; values sent locally only)

A10.8.1.8. 6 - Weather and obstruction to vision (if any)

A10.8.1.9. 7 - Sky condition

A10.8.1.10. 8 - Temperature and dew point

A10.8.1.11. 9 - Altimeter setting

A10.8.1.12. 10 - Remarks (as required) **Note**: AO2A signifies that the observations was supplemented

A10.8.1.13. 11 -Minutes after the hour/initials

A10.9. Special Observation (SPECI) Format (Supplemented Observation).

A10.9.1. - (1)(2)(3)(4)(5)(6)(7)(8)(9)

A10.9.1.1. KBIXSPECI1342ZVRB04KT3/4SMR04/1600FTBRBKN005 15/14A2992 RMK AO2A PRESFR SLP057 $ PA +7 DA +2263 43/BF

A10.9.1.2. - (10)(11)

A10.9.1.3. 1 - Station identifier and observation type

A10.9.1.4. 2 - Time

A10.9.1.5. 3 - Wind direction and speed

A10.9.1.6. 4 - Visibility

A10.9.1.7. 5 - Runway visual range (when required; values sent locally only)

A10.9.1.8. 6 - Weather and obstruction to vision (if any)

A10.9.1.9. 7 - Sky condition

A10.9.1.10. 8 - Temperature and dew point

A10.9.1.11. 9 - Altimeter setting

A10.9.1.12. 10 - Remarks (as required) **Note**: AO2A signifies that the observations was supplemented

A10.9.1.13. 11 -Minutes after the hour/initials

A10.10. PIREP Format.

A10.10.1. **-** (1)(2)(3)(4)(5)(6)(7)

A10.10.1.1. KBIXPIREPTIME1500KGPT FL 040 TP C130SK 050 SCT UNKNWX R-

A10.10.1.2. TA 10TURB LGT ICE NEG

A10.10.1.3. **-** (8)(9)(10)

A10.10.1.4. 1 - Station identifier

A10.10.1.5. 2 - Time of the report (Z time)

A10.10.1.6. 3 - Location of aircraft

A10.10.1.7. 4 - Flight level

A10.10.1.8. 5 - Type aircraft

A10.10.1.9. 6 - Sky condition

A10.10.1.10. 7 - Weather at flight level

A10.10.1.11. 8 - Outside air temperature in Celsius

A10.10.1.12. 9 - Turbulence reported (if any)

A10.10.1.13. 10 - Icing reported (if any)

Attachment 11

KEESLER AFB SPECIAL OBSERVATION CRITERIA

A11.1. Special Observations. The Weather Forecaster takes a special observation (SPECI) to report weather conditions passing through critical threshold values important to flying operations. BOS/WSO disseminates these observations via the ADS. With the exception of single element specials, these observations contain the following information: date/time, sky condition, prevailing visibility, present weather and obstruction to visibility (if required), temperature, dew point, wind speed and direction (magnetic), altimeter setting, remarks (as appropriate), and RVR.

A11.2. Criteria for Special Observations.

A11.2.1. Ceilings. A ceiling forms or dissipates below, decreases to less than or, if below, increases to equal or exceeds:

Table A11.1. Ceilings.

3,000 feet (AFMAN 15-111)	600 feet (DoD FLIP)
2,000 feet (AFMAN 15-111)	500 feet (AFMAN 15-111) (DoD FLIP)
1,500 feet (AFMAN 15-111)	300 feet (AFMAN 15-111)
1,000 feet (AFMAN 15-111) (DoD FLIP)(403 WG)	200 feet (DoD FLIP)
800 feet (AFMAN 15-111) (DoD FLIP)	100 feet (AFMAN 15-111)
700 feet (AFMAN 15-111) (DoD FLIP)	

A11.2.2. Prevailing Visibility**.** The prevailing visibility decreases to less than, or if below, increases to equal or exceeds:

Table A11.2. Visibilities.

3 miles (AFMAN 15-111)	1 mile (AFMAN 15-111)(DoD FLIP)(403 WG)
2 1/2 miles (DoD FLIP)	3/4 mile (AFMAN 15-111) (DoD FLIP)
2 miles (AFMAN 15-111)(DoD FLIP)	1/2 mile (AFMAN 15-111) (DOD Flip)
1 3/4 miles (DoD FLIP)	1/4 mile (AFMAN 15-111)
1 5/8 miles (403 WG)	
1 1/2 miles (AFMAN15/111)	
1 3/8 miles (DoD FLIP)	

A11.2.3. Sky Condition.

A11.2.3.1. A layer of clouds or obscuring phenomena aloft observed below **800 feet** and no layer aloft was reported below **800 feet** in the preceding observation.

A11.2.4. Wind Shift.

A11.2.4.1. A Change in wind direction of 45 degrees or more in less than 15 minutes when wind speeds are 10 knots or more throughout the wind shift.

A11.2.5. Squall.

A11.2.5.1. When occurs.

A11.2.5.2. Defined: A strong wind characterized by a sudden onset, in which wind speed increases at least 16kts and is sustained at 22 kts or more for at least 1 minute, followed by a sudden decrease in speed.

A11.2.6. Volcanic Eruption**.**

A11.2.6.1. Eruption or volcanic ash cloud first observed.

A11.2.7. Thunderstorm.

A11.2.7.1. Begins (not required if one is currently reported as in progress at the station). **Note**: Considered to have begun when thunder is first heard. Also considered when hail is falling, lightning is observed near the airfield, **and** local noise level is such that thunder cannot be heard.

A11.2.7.2. Ends. **Note**: Considered to have ended 15 minutes after the last occurrence of criteria above.

A11.2.8. Precipitation**.**

Table A11.3. Precipitation.

Precipitation Type	**Criteria**
Hail	Begins or ends
Freezing	Begins, ends, or changes intensity
Ice Pellets	Begins, ends, or changes intensity
All others	Begins or ends
Note: SPECI is **not required** for changes in type or the beginning or ending of one type while another is in progress **except freezing rain, freezing drizzle, hail, and ice pellets**.	

A11.2.9. Tornado (+FC), FUNNEL CLOUD (FC), OR WATERSPOUT (+FC)

A11.2.9.1. Is observed.

A11.2.9.2. Disappears from sight or ends.

A11.2.10. RVR.

A11.2.10.1. Report RVR when the Prevailing Visibility is First observed <1SM, or if below, rises above 1SM.

A11.2.10.2. RVR values decrease to less than or, if below, increases to equal or exceed (Active runway only):

Table A11.4. RVR.

6,000 feet	(AFMAN 15-111)	2,000 feet	(AFMAN 15-111)
5,500 feet	(DoD FLIP)	1,600 feet	(AFMAN 15-111)
5,000 feet	(AFMAN 15-111)	1,200 feet	(AFMAN 15-111)
4,500 feet	(DoD FLIP)	1,000 feet	(AFMAN 15-111)
4,000	(AFMAN 15-111)(DoD FLIP)	600 feet	(AFMAN 15-111)
2,400 feet	(AFMAN 15-111) (DoD FLIP)	At the request of aircrews or Tower **and** value differs from the previously reported value.	

A11.2.11. During periods of back up operations, due to a total loss of the AN/FMQ-19 AMOS, the following criteria will necessitate a SPECI observation to be taken:

A11.2.11.1. Runway Change.

A11.2.11.1.1. Take and disseminate an observation when notified of a change in runway. Time of observation is the time the last element is observed, **not** time of notification.

A11.2.11.2. Altimeter Settings.

A11.2.11.2.1. Only disseminated during periods the airfield is open, and ATC does not have access to real-time ALSTGs.

A11.2.11.2.2. Taken at an interval not to exceed 35 minutes when there has been a change of 0.01 inch Hg (0.3 hPa) or more since the last ALSTG value. (**Note**: A METAR or SPECI taken within the established time interval will meet this requirement.

A11.2.11.3. Disaster Response Force (DRF) Activation/Toxic Corridor.

A11.2.11.3.1. Immediately following notification of an on or off base accident requiring a Toxic Corridor or activation of the DRF.

A11.2.11.4. AN/FMQ-19 AMOS Outage or Erroneous Readings.

A11.2.11.4.1. Taken during periods the airfield is open and both sensors are out of service or suspected of erroneous readings.

A11.2.11.4.2. The forecaster will take single element wind SPECI, using the Kestrel, at 15, 30, and 45 minutes past each hour, unless intervened by a SPECI, until the AN/FMQ-19 AMOS is deemed reliable. **Note**: Include WND DATA ESTMD in remarks.

A11.2.11.5. Inactive Runway Wind Data – During Backup Operations Only.

A11.2.11.5.1. When a difference in wind speed between the active and inactive sensor is >10kts (including gusts).

A11.2.11.6. Aircraft MISHAP.

A11.2.11.6.1. Upon notification or sighting, at or near the station, of an Aircraft Mishap, the Weather Forecaster will take and record a full element SPECI observation (print a copy of the 3813 and write the remark “ACFT MISHAP” in Col 13, but don’t transmit the remark).

A11.2.11.7. Upon Resumption of Services.

A11.2.11.7.1. Within 15 minutes of returning for duty following a break in hourly coverage.

A11.2.11.7.2. When opening at the time of an hourly, transmit a METAR.

A11.2.11.8. When the tower reports.

A11.2.11.8.1. the visibility has decreased to less than or, if below, increased to equal or exceed 1, 2, or 3 statute miles, and differs from the prevailing visibility

A11.2.11.8.2. the visibility has decreased to less than or, if below, increased to equal or exceed tower special criteria and differs from the prevailing visibility

Attachment 12

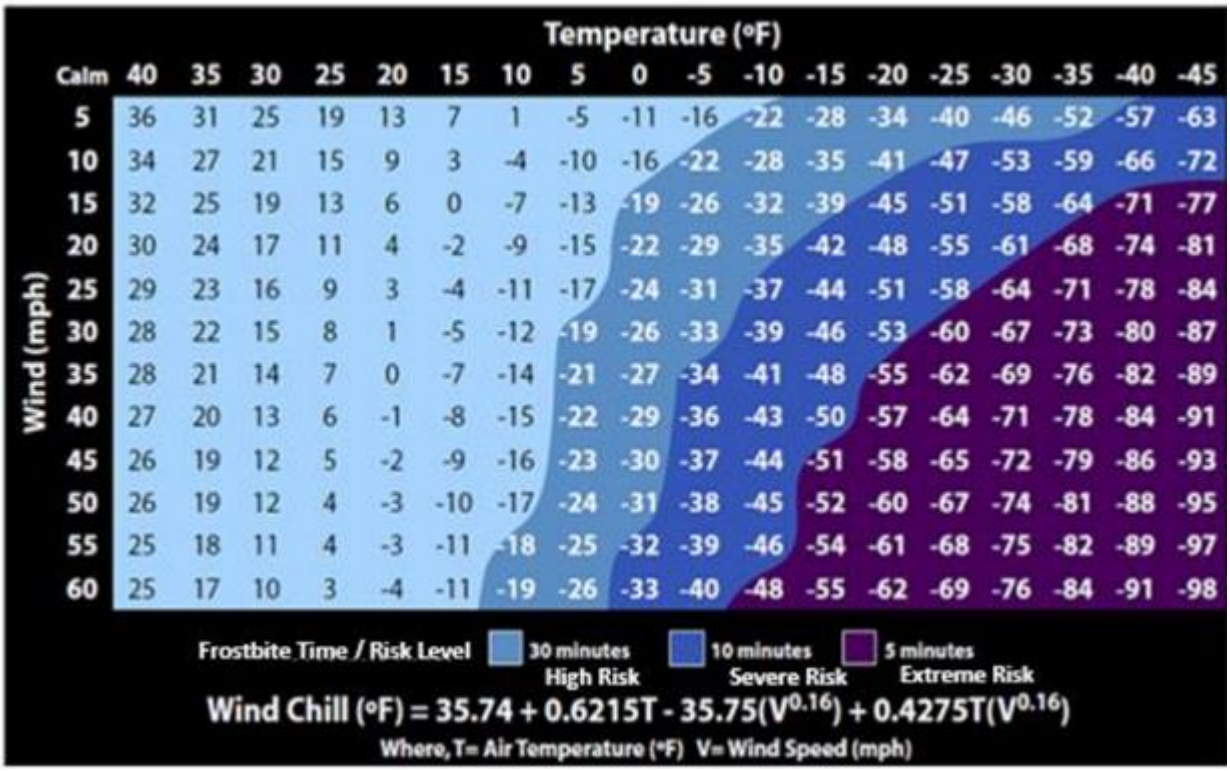
INDEX OF THERMAL STRESS/ CHILL INDEX CATEGORY CHARTS

A12.1. Index of Thermal Stress.

Table A12.1. Dewpoint 23C to 46C.

Dry Bulb Temp (°C)	Zone	Dewpoint (°C)																						
		23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46
23		32	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
24		33	33	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
25		33	34	35	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
26		34	35	35	36	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
27	CAUTION	35	35	36	36	37	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
28		35	36	36	37	38	38	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
29		36	36	37	38	38	39	39	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
30		36	37	38	38	39	39	40	41	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
31		37	38	38	39	39	40	40	41	42	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
32		38	38	39	39	40	40	41	42	42	43	X	X	X	X	X	X	X	X	X	X	X	X	X	X
33		38	39	39	40	40	41	42	42	43	43	44	X	X	X	X	X	X	X	X	X	X	X	X	X
34		39	39	40	40	41	41	42	43	43	44	45	45	X	X	X	X	X	X	X	X	X	X	X	X
35		39	40	40	41	41	42	43	43	44	45	45	46	46	X	X	X	X	X	X	X	X	X	X	X
36		40	40	41	41	42	43	43	44	44	45	46	46	47	48	X	X	X	X	X	X	X	X	X	X
37		40	41	41	42	43	43	44	44	45	46	46	47	47	48	49	X	X	X	X	X	X	X	X	X
38		41	41	42	42	43	44	44	45	45	46	47	47	48	49	49	50	X	X	X	X	X	X	X	X
39		41	42	42	43	44	44	45	45	46	47	47	48	49	49	50	51	51	X	X	X	X	X	X	X
40	DANGER	42	43	43	44	44	45	45	46	46	47	48	48	49	50	50	51	52	52	X	X	X	X	X	X
41		43	43	44	44	45	45	46	46	47	48	48	49	50	50	51	52	52	53	54	X	X	X	X	X
42		43	44	44	45	45	46	46	47	48	48	49	49	50	51	51	52	53	53	54	55	X	X	X	X
43		44	44	45	45	46	46	47	47	48	49	49	50	51	51	52	52	53	54	55	55	56	X	X	X
44		44	45	45	46	46	47	47	48	49	49	50	50	51	52	52	53	54	54	55	56	56	57	X	X
45		45	45	46	46	47	47	48	48	49	50	50	51	52	52	53	53	54	55	55	56	57	58	58	X
46		45	46	46	47	47	48	48	49	50	50	51	51	52	53	53	54	55	55	56	57	57	58	59	60

Table A12.2. Chill Index (Index Chart for 33-foot Anemometer).



Wind (mph) \ Temperature (°F)	40	35	30	25	20	15	10	5	0	-5	-10	-15	-20	-25	-30	-35	-40	-45
Calm																		
5	36	31	25	19	13	7	1	-5	-11	-16	-22	-28	-34	-40	-46	-52	-57	-63
10	34	27	21	15	9	3	-4	-10	-16	-22	-28	-35	-41	-47	-53	-59	-66	-72
15	32	25	19	13	6	0	-7	-13	-19	-26	-32	-39	-45	-51	-58	-64	-71	-77
20	30	24	17	11	4	-2	-9	-15	-22	-29	-35	-42	-48	-55	-61	-68	-74	-81
25	29	23	16	9	3	-4	-11	-17	-24	-31	-37	-44	-51	-58	-64	-71	-78	-84
30	28	22	15	8	1	-5	-12	-19	-26	-33	-39	-46	-53	-60	-67	-73	-80	-87
35	28	21	14	7	0	-7	-14	-21	-27	-34	-41	-48	-55	-62	-69	-76	-82	-89
40	27	20	13	6	-1	-8	-15	-22	-29	-36	-43	-50	-57	-64	-71	-78	-84	-91
45	26	19	12	5	-2	-9	-16	-23	-30	-37	-44	-51	-58	-65	-72	-79	-86	-93
50	26	19	12	4	-3	-10	-17	-24	-31	-38	-45	-52	-60	-67	-74	-81	-88	-95
55	25	18	11	4	-3	-11	-18	-25	-32	-39	-46	-54	-61	-68	-75	-82	-89	-97
60	25	17	10	3	-4	-11	-19	-26	-33	-40	-48	-55	-62	-69	-76	-84	-91	-98

Frostbite Time / Risk Level: 30 minutes High Risk; 10 minutes Severe Risk; 5 minutes Extreme Risk

$$\text{Wind Chill (°F)} = 35.74 + 0.6215T - 35.75(V^{0.16}) + 0.4275T(V^{0.16})$$

Where, T= Air Temperature (°F) V= Wind Speed (mph)